Final Year Design Project

Final Project Report for

Tracefy: Generating sketches and face images of perpetrators for efficient identification

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Abstract

Tracefy is an advanced generative AI system that transforms hand-drawn or digitally outlined facial sketches into high-quality, realistic facial images guided by textual descriptions. This project addresses the gap between artistic input and photorealistic image generation using a multi-stage deep learning pipeline, combining image translation, prompt-conditioning, and guided diffusion models. The system architecture integrates a CycleGAN-based sketch refinement module, followed by a dual-conditioning image generation pipeline utilizing a fine-tuned FLUX.1-dev latent diffusion model and its corresponding ControlNet extension (FLUX.1-dev-ControlNet-Union-Pro-2.0).

The pipeline accepts a facial outline and natural language input from the user. The sketch is refined to improve anatomical accuracy and structural coherence, then used along with a descriptive prompt to condition the ControlNet module, ensuring spatial fidelity, while the text prompt shapes semantic and aesthetic details through the base diffusion model. A custom lightweight LoRA fine-tuning mechanism is employed to enhance generation quality on identity preservation and sketch alignment.

Evaluation of generated images is performed using a combination of perceptual and structural metrics, including Canny edge detection for edge structure comparison, perceptual hash (pHash) similarity for visual similarity assessment, and Hamming distance to quantify hash differences between generated and reference images.

The user interface is designed as a web-based application offering intuitive sketching, prompt input, and real-time image generation. From a business standpoint, Tracefy serves creative professionals, forensic artists, and entertainment designers, offering a scalable platform for personalized image generation. Stakeholders include AI researchers, product designers, digital artists, and law enforcement agencies. The project lays a foundation for future work in expanding modality input types, improving personalization, and deploying in mobile environments.

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Revision History

Name	Date	Reason for changes	Version

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1. Introduction

1.1. Purpose

The following document serves as a comprehensive guide to the project Tracefy: Generative AI-based face generation system. It is intended as the primary source for each requested feature of the system, the rationale behind the functionality of each feature, and other critical aspects of the product's design plan required before and during the product's implementation. A quick note about reading this document: it is not intended to be read sequentially from start to finish. Instead, it is recommended to navigate through the document to locate the specific information you need. For ease of reference, a detailed table of contents is provided on page 1.

1.2. Product Perspectives

Tracefy is an innovative, standalone system designed to generate accurate criminal sketches based on visual attributes provided by witnesses or victims. Developed to support law enforcement agencies in Pakistan, Tracefy enhances the efficiency and effectiveness of criminal investigations. Unlike follow-on products or system replacements, Tracefy introduces a novel AI-driven approach using advanced technologies such as Latent Diffusion Models (LDMs) and Generative Adversarial Networks (GANs) to produce highly detailed sketches. While it operates independently, Tracefy is built to integrate seamlessly with existing criminal databases, case management systems, and other law enforcement tools.

1.3. Overview of Criminal Activities in Pakistan

Criminal activities continue to pose a serious threat to global peace and public security, compelling nations to adopt proactive measures to curb the escalation of unlawful behavior. These activities span a broad spectrum, including violent crimes such as homicide and property crimes like robbery. According to the Global Organized Crime Index, as of 2023, approximately 83% of the world's population lives in countries experiencing high levels of criminality. Pakistan, being the fifth most populous country and still navigating the challenges of development, is not immune to the surge of urban crimes, particularly in its major metropolitan areas. A combination of factors—including rising inflation, widespread unemployment, recurring food shortages, environmental calamities, and persistent political instability—has significantly impacted the socio-economic landscape. These pressures have pushed a substantial portion of the youth population below the poverty line, making them more vulnerable to recruitment into criminal networks. A grim example is presented by the Karachi Police, who reported a staggering 60,580 street crime incidents within the first nine months of 2022 alone. Notably, 33.33% of these incidents involved mobile phone snatching (Ayaz, 2023). Such statistics paint a concerning picture of law and order in urban centers and highlight the urgent need for innovative, technology-driven solutions to combat crime and promote safer living conditions for citizens across the country.

1.4. Importance of Generative Models

Generative models have emerged as a transformative advancement in artificial intelligence, enabling machines to create new, realistic data rather than merely analyzing existing information. Their importance lies in their ability to synthesize high-quality images, text, audio, and even 3D content based on learned patterns from training data. In domains such as design, medicine, and law enforcement, generative models offer unprecedented creative assistance, automation, and reconstruction capabilities. Specifically in systems like Tracefy, they bridge the gap between human imagination and visual realism—turning rough sketches and abstract descriptions into detailed, lifelike facial images—thereby enhancing both productivity and accessibility in traditionally manual, expert-driven workflows.

1.5. Motivation

Tracefy was developed to address fundamental flaws in Pakistan's criminal justice system, where high crime rates and growing case backlogs often stem from a lack of usable suspect information. The problem goes beyond technical limitations—it includes emotional, social, and bureaucratic challenges. Public trust declines when justice is delayed or fails, especially in high-profile cases involving gender-based violence and child abductions.

Tracefy fills this gap by using AI to generate facial sketches directly from victim or witness descriptions, unlike existing systems that rely on CCTV footage or prior imagery. It integrates with current digital initiatives such as e-challans, digital FIRs, and the Face Trace System, but focuses specifically on generating images from recollections. By enabling rapid transformation of text descriptions into facial visuals, Tracefy speeds up suspect identification, optimizes law enforcement resources, and supports victims with faster progress updates. It reduces reliance on inaccurate, opinion-based sketches and offers standardized, reliable outputs. Ultimately, Tracefy strengthens investigative capabilities and lays the foundation for future predictive policing systems.

1.6. Objectives of the Project

Tracefy represents an advanced technological initiative designed to modernize Pakistan's criminal justice system by leveraging data-driven and AI-powered solutions to address technical, operational, and societal challenges. The project is built around the following key objectives:

1.6.1. Modernize Suspect Identification

Tracefy aims to establish a standardized and reliable method for generating realistic facial images using AI, based on textual descriptions provided by victims or witnesses. By overcoming the limitations of traditional hand-drawn sketches, this system modernizes investigative techniques and supports more precise suspect profiling.

1.6.2. Enable Real-Time Image Generation and Matching

The system is designed to generate suspect visuals in near real-time and facilitate their comparison against existing records in national databases such as NADRA. This functionality supports faster identification and investigation, increasing the chances of timely arrests and reducing case backlogs.

1.6.3. Develop a Continuously Learning Criminal Database

Tracefy includes a self-adaptive database system that evolves with each use. Every investigative query contributes to expanding and refining the dataset, thereby enhancing the system's accuracy and effectiveness in future identification tasks.

1.6.4. Strengthen Institutional Collaboration

The system supports secure integration with government and law enforcement infrastructure. It facilitates seamless data exchange between police, intelligence, and prosecution departments, enabling a cohesive and coordinated approach across the criminal justice chain.

1.6.5. Minimize Resource Dependence and Human Error

By reducing the reliance on scarce and costly forensic sketch artists, Tracefy provides a more accessible and consistent alternative. The AI-driven process minimizes subjective interpretation and human error, freeing valuable human resources for other critical investigative work.

1.6.6. Support Time-Critical Investigations

In high-priority cases such as child abductions, missing persons, or serial offenses, Tracefy empowers law enforcement agencies to act swiftly and decisively. The system's efficiency in generating suspect visuals can be crucial in situations where delays may result in irreversible outcomes.

1.6.7. Alleviate Emotional Burden on Victims

By standardizing and simplifying the suspect identification process, Tracefy reduces the cognitive and emotional strain placed on victims. It offers a more compassionate alternative that allows victims to participate in investigations without the distress of detailed memory recall or lengthy interviews.

1.6.8. Enhance Public Trust in Law Enforcement

Through greater accuracy, transparency, and faster case resolution, Tracefy aims to restore public confidence in law enforcement agencies. Its visible impact on improving suspect identification reinforces institutional credibility and accountability.

1.6.9. Promote Sustainable Innovation in Public Safety

Aligned with the United Nations Sustainable Development Goals (SDGs) 9 and 16, Tracefy contributes to national progress through innovation in digital infrastructure and supports the development of inclusive, fair, and effective justice institutions.

2. Background and Literature Review

2.1. Problem Statement

Criminal sketches drawn by humans are slow, inaccurate and riddled with bias, leading to low apprehension rates. The application will aim to solve this problem as it will provide speed, accuracy and impartiality. With the fulfilment of these factors, this implementation can be catered to the government. As a B2G category product, it will fill the lack of modern tools and increase apprehension rates, thereby decreasing crime.

2.2. Literature Review

The literature review for this project is based on the existing work related to the idea of our project and details about the implementation that we expect to follow. These papers provide useful insight on what types of datasets we can use, and the limitations and advantages of the techniques used in generative-AI technology.

2.2.1. Face Synthesis from Visual Attributes via Sketch using Conditional VAEs and GANs (Xing Di, Vishal M. Patel (2017))

This paper introduces the **Attribute2Sketch2Face** framework, a three-stage generative process involving Conditional Variational Autoencoders (CVAEs) and GANs. It begins with generating rough sketches from visual attributes, refines these sketches using GAN-based enhancement, and finally produces realistic colored facial images.

- **Relevance to Tracefy**:
  The staged approach directly inspires Tracefy's architecture, particularly the breakdown into Attribute-to-Sketch (A2S), Sketch-to-Sketch (S2S), and Sketch-to-Face (S2F) components. It validates the effectiveness of intermediate sketch generation for managing complexity and improving interpretability.
- **Limitations**:
  - Heavy reliance on biased datasets like CelebA and LFW, lacking diversity and representativeness.
  - High computational cost despite a staged approach.
  - No deployment considerations or integration with real-world investigative systems.
  - Generated outputs may lack forensic-grade precision for legal admissibility.

2.2.2. A Sketch-Based Image Retrieval Technique (Abdulaali, Sulong, Abdulbaqi, Hashem (2014)))

This paper proposes a method for retrieving images based on input sketches using feature extraction techniques like SIFT, ShoG, and Bag of Words. It achieves strong precision and recall metrics by matching sketch features to indexed database images.

- **Relevance to Tracefy**:
  Forms the basis of Tracefy’s sketch-based image retrieval (SBIR) module. The approach enables suspect identification by matching user-generated sketches to mugshot or surveillance image databases.
- **Limitations**:
  - Struggles with abstract or imprecise sketches.
  - Performance degrades without skilled input sketches.
  - Limited support for color and texture features.
  - Relies on traditional feature extractors rather than deep learning methods.

2.2.3. High-Resolution Image Synthesis with Latent Diffusion Models (Rombach, Blattmann, Lorenz, Esser, Ommer (2021))

This work introduces Latent Diffusion Models (LDMs), which generate high-resolution images by denoising in a compressed latent space, reducing computational load while maintaining visual quality.

- **Relevance to Tracefy**:
  LDMs are integrated into Tracefy's architecture for both sketch generation and enhancement. Their latent-space operations offer efficient and scalable high-quality synthesis—crucial for generating detailed forensic sketches.
- **Limitations**:
  - Training complexity and resource demands remain significant.
  - Over-compression in latent space can reduce fine detail accuracy.
  - Dependent on training dataset quality and diversity.

2.2.4. Removing Biased Data to Improve Fairness and Accuracy (Sahil Verma, Michael Ernst, Rene Just (2021))

The paper outlines techniques like Exploratory Data Analysis (EDA), AIF360, Adversarial Debiasing, and Regularization Penalties to detect and mitigate bias in machine learning models.

- **Relevance to Tracefy**:
  Tracefy incorporates these fairness-aware mechanisms to avoid biased outputs in sketch generation, especially related to sensitive attributes like race, gender, and age.
- **Limitations**:
  - Debiasing methods can be complex to tune.
  - Fairness trade-offs must be balanced carefully to preserve output fidelity.

2.2.5. Forensic Sketch AI-rtist (Existing System) (Artur Fortunato & Filipe Reynau)

This system uses DALL·E to generate facial images from textual descriptions. It showcases the use of large text-to-image models in forensic applications.

- **Relevance to Tracefy**:
  Highlights the potential of large-scale generative models in suspect sketching and motivates Tracefy's exploration of similar generation capabilities—but with more control and interpretability.
- **Limitations**:
  - Biased output due to opaque pretraining data.
  - Lack of grounding in visual sketch control leads to mismatches.
  - Insufficient precision for real-world legal and forensic standards.

2.3. Summary of Literature Review

Existing System	**Key Contribution**	**Limitations Overcome by Tracefy**
Attribute2Sketch2Face	Modular generative sketch-to-image pipeline	Adds fairness, LDM efficiency, and improved real-world focus
Sketch-Based Retrieval	Accurate matching via feature descriptors	Replaces hand-crafted features with deep, robust embeddings
Latent Diffusion Models	High-resolution generation in latent space	Adapted for sketches and conditional generation
Fairness Frameworks	Practical bias mitigation techniques	Integrated to ensure ethically sound outputs
Forensic Sketch AI-rtist	Text-to-image via DALL·E	Replaced with sketch-guided, controlled generation pipeline

Table 1: Summary of Literature Review

3. Design Methodology and Software Process Model

The project adopts an Agile-based iterative development model to enable ongoing refinement and effective collaboration. This section details the design approach, development stages, and the reasoning behind selecting this software process model, emphasizing its suitability for ensuring timely and adaptable delivery.

3.1. Design Methodology: Modular Approach

A modular design approach was adopted for this project to support flexibility, maintainability, and ease of experimentation. This decision was guided by the nature of the tasks involved, which—though sequential—benefit from being implemented and optimized independently. The system was divided into distinct functional units, each responsible for a specific stage in the pipeline: dataset preparation, sketch refinement, image generation, and system integration.

Each module was designed to operate autonomously, with clearly defined inputs and outputs. For example, the sketch refinement module transforms rough user input into clean line art using a trained CycleGAN model. This output is then passed to the controlled image generation module, where a combination of ControlNet and Stable Diffusion creates realistic face images based on both the refined sketch and a descriptive prompt.

This modular separation allows for parallel development, simplified debugging, and future scalability. Updates or improvements to one module—such as swapping in a different image-to-image model—can be made without affecting the functionality of others. The final integration was achieved through a full-stack web application, where frontend and backend components were connected via RESTful APIs, ensuring smooth communication between modules and a responsive user interface.

3.2. Reasons for choosing Modular Approach

The following reasons make it compelling to choose modular approach for Tracefy:

3.2.1. Parallel Development

A modular architecture allowed different team members to work on distinct components simultaneously. For instance, while one part of the team focused on training CycleGAN for sketch refinement, others could implement the user interface or integrate ControlNet for image generation—significantly accelerating development time.

3.2.2. Ease of Testing and Debugging

Since modules were self-contained, individual components could be tested and debugged in isolation. This made it easier to identify bottlenecks or failures without having to step through the entire pipeline, thereby improving development efficiency and reliability.

3.2.3. Scalability and Flexibility

The modular design made it possible to replace or upgrade components without rewriting the entire system. For example, future iterations could incorporate a different sketch-to-image model or a more advanced prompt processing technique with minimal disruption to the overall architecture.

3.2.4. Reusability and Integration

Modules developed for Tracefy can be reused in other AI-driven applications involving image translation, text conditioning, or generative models. Moreover, the clear boundaries between modules allowed seamless integration via REST APIs, making the system compatible with modern full-stack web frameworks and deployment platforms.

3.2.5. Limitations of OOP

Object-oriented programming introduces additional complexity despite offering modularity and reusability, especially in tasks like machine learning pipelines where interactions between objects are minimal. While OOP may be less critical in such scenarios, it becomes more beneficial for larger projects that involve multiple interacting components.

3.3. Software Process Model: Incremental Approach

The Incremental Model is the software process model selected for the Tracefy project. The iterative nature of developing an AI-based sketch and text-guided image generation system, combined with the need for continuous improvements in model performance and user interface, made this model the most suitable choice.

3.3.1. Reasons for choosing Incremental Model

The core reason for selecting the incremental and iterative model lies in the experimental and modular structure of Tracefy. Since Tracefy integrates advanced generative AI techniques, user interface components, and performance-tuning mechanisms, it benefits significantly from a development strategy that supports:

- Gradual feature addition (incremental development)
- Continuous refinement through feedback (iterative enhancement)
- Early and frequent delivery of functional components
- Risk reduction through isolated testing of individual modules (e.g., sketch processing, text interpretation, image synthesis)

3.3.2. Incrementation Details

- **Initial Iteration – Core Functionality**

The first iteration focused on building a minimal viable system that could accept sketch and text inputs to generate basic image outputs. This included a rough image synthesis pipeline using pre-trained models and a simple UI for interaction.

- **Subsequent Increments – Enhancements and Additions**

Additional increments introduced:
- Enhanced sketch-to-image quality through fine-tuning
- Natural language prompt conditioning
- Modular integration of upscaling and post-processing steps
- UI improvements based on user testing

- **Iterative Refinement – Based on Feedback and Evaluation**

Each module (e.g., image generation model, sketch interpreter, prompt parser) undergoes iterative tuning based on:
- Qualitative user feedback
- Performance metrics (e.g., realism, fidelity, response time)
- Usability testing results
- Parallel Iteration of Components

The backend ML model and frontend UX are developed in parallel iterations, allowing fast prototyping of new ideas without waiting for the full system to stabilize.

3.3.3. Benefits of Using Incremental Model

Flexibility to Adapt to User Needs as the System Evolves

Tracefy's user base includes artists, designers, and developers who may have varying expectations regarding the quality and style of generated images. By following the incremental and iterative model, the development team can quickly incorporate user feedback into future iterations. For example, if users request more realistic facial rendering or support for different sketching styles, these changes can be prioritized and added in a focused update without overhauling the entire system. This flexibility ensures that Tracefy remains responsive to changing user needs and emerging trends in generative AI.

Faster Detection of Design and Usability Issues

Since the system is built and tested in small increments, potential design flaws or usability bottlenecks are identified early in the development process. Each iteration provides a testable version of the system, allowing developers to observe user behavior and collect feedback before investing time into full-scale implementation. For instance, if users struggle to understand how to use the sketch upload tool, the issue can be corrected in the next cycle, reducing rework and improving the overall user experience.

Efficient Integration of Research-Oriented Experiments

Tracefy relies heavily on experimentation with different AI models and image generation techniques. The iterative model supports this by allowing the team to isolate and test new algorithms—such as prompt-to-style mapping, image enhancement, or multimodal fusion—without disrupting the core functionality. Successful experiments can be gradually integrated into the production pipeline, while less effective ones can be discarded early, saving resources and development time.

Scalable Architecture, Where Each Module Can Be Enhanced Independently

The modular structure encouraged by the incremental model makes Tracefy's architecture inherently scalable. Each major component—such as the sketch interpreter, text encoder, image generator, and UI—can be developed and refined independently. This means that improving the image synthesis engine (e.g., swapping to a newer model or adding a denoising step) doesn't require rebuilding the entire system. This separation of concerns also enables parallel development across teams and simplifies future maintenance and upgrades.

4. System Overview

Tracefy is an advanced, AI-driven image synthesis platform designed to generate high-fidelity images conditioned on dual-modality inputs—specifically, user-supplied sketches and natural language prompts. The core objective of the system is to enable intuitive and controllable image generation by combining minimal visual cues with descriptive linguistic context. This dual-conditioning paradigm allows users to maintain fine-grained control over image structure (via sketch) and semantic style or content (via text), which is particularly useful in creative workflows such as digital art, concept design, and rapid prototyping.

4.1. System Design

The design of Tracefy is grounded in a human-centric philosophy that prioritizes intuitive usability, expressive freedom, and technological robustness. The system is intended to serve as a creative partner, not just a technical tool.

4.1.1. Core Design Principles:

- **Imagination-first Interaction**: Users can express their ideas through rough sketches and simple language, rather than needing advanced artistic or technical skills.
- **Semantic-Visual Fusion:** Tracefy excels at blending the what (textual traits) and the how (sketched form) of user intent into a single, coherent image.
- **Respect for User Intent:** Each module in the system is tuned to preserve and enhance the user's creative vision, rather than override it with generic outputs.
- **Modularity and Scalability:** The pipeline is modular, making it easy to upgrade components (e.g., switching sketch refiners or text encoders) and expand capabilities in future versions.

Tracefy ultimately serves as a bridge between abstract imagination and concrete visual expression, unlocking new possibilities in digital creativity, education, design, and applied AI.

4.2. System Objectives

- **Sketch-to-Image Translation**: Generate realistic face images from simple user-drawn outlines.
- **Textual Guidance**: Incorporate semantic details (e.g., age, hair color, emotions) from descriptive text prompts.
- **Realism and Precision**: Produce high-fidelity facial imagery with strong adherence to user input.
- **Accessibility**: Offer a seamless and intuitive experience to users without requiring artistic or technical expertise.

4.3. System Modules

4.3.1. User Input Module

The User Input Module is the initial interaction point between the user and the Tracefy system. It is designed to be intuitive and accessible, catering to both technically adept users and non-technical creatives such as artists, educators, and designers. This module accepts two primary types of user input:

Input Type 1: Sketch Input

The users can upload a line-art sketch. This sketch typically includes key facial outlines and contours — such as the shape of the head, placement of the eyes, eyebrows, nose, mouth, and the general hairline. The input is not expected to be artistically perfect; instead, it serves as a rough blueprint or scaffold that conveys spatial information about facial structure and orientation. This visual input allows the user to express their intended facial geometry in a quick and intuitive manner.

Input Type 2: Textual Prompt
Accompanying the sketch is a natural language description, where users articulate the desired attributes of the face in words. This prompt can include a variety of descriptive elements such as:

- Demographic details: age, gender, ethnicity
- Physical attributes: hairstyle, hair color, skin tone, eye shape, or presence of facial marks
- Emotional state or expression: smiling, angry, thoughtful, surprised, etc.
- Accessories and adornments: glasses, earrings, headwear, facial piercings, etc.

The textual input enables users to communicate complex, high-level features and abstract traits that are difficult to capture in a simple sketch, thus enriching the generative process with semantic depth.

4.3.2. Sketch Refinement Module

Given that user sketches are often imprecise or inconsistent — especially when drawn freehand or using touch interfaces — the Sketch Refinement Module serves a critical preprocessing role. This module is responsible for converting the rough sketch into a cleaner, more coherent representation that is suitable for computational processing in subsequent stages.

Model Used: Cycle GAN (Image-to-Image Translation)
The system employs a Cycle GAN-based architecture for image-to-image translation. Cycle GAN is particularly well-suited for unpaired data translation, making it ideal for transforming rough human-drawn sketches into refined line drawings without requiring exact paired ground truth samples.

Purpose and Functionality
The primary function of this module is to:

- Remove inconsistencies and artifacts from the hand-drawn sketch.
- Enhance edge definitions and facial structure.
- Standardize the drawing to meet the expectations of downstream edge detection algorithms.

The output of the sketch refinement process is a clean, high-contrast facial outline that accurately preserves the user's intent while offering a more stable and interpretable visual structure. This refined sketch plays a foundational role in defining the spatial constraints during image generation.

4.3.3. Image Generation Module

The Image Generation Module is the core of the Tracefy system, where the final face image is synthesized based on the encoded inputs. This stage integrates advanced diffusion-based generative modeling with spatial and semantic control mechanisms to produce realistic and contextually accurate results.

Base Model: FLUX.1-dev
Tracefy employs FLUX.1-dev, a diffusion-based latent generative model that is particularly well-suited for high-fidelity face generation. FLUX.1-dev has been optimized for expressive control and fine-detail rendering, making it an excellent backbone for this application.

Control Mechanism: ControlNet-Union-Pro-2.0
To ensure that the final image adheres to the user-provided sketch structure, ControlNet-Union-Pro-2.0 is used as the conditioning mechanism. ControlNet enables the diffusion model to incorporate external constraints — such as edge maps — directly into the generation process, preserving facial proportions, alignment, and geometry as defined by the user's input.

Fine-Tuning Strategy: LoRA Adapters
Tracefy incorporates Low-Rank Adaptation (LoRA) modules to fine-tune the generative model without full retraining. LoRA adapters enable efficient and targeted adaptation to specific visual styles, facial diversity, or prompt domains, offering flexibility and personalization within the generation pipeline.

This module produces a high-resolution, photorealistic image that successfully merges the spatial fidelity of the input sketch with the semantic richness of the text description. The resulting face exhibits lifelike texture, emotion, and distinctiveness, often evoking a sense of realism and character.

4.4. Apportioning of Requirements

Frontend and Backend Responsibilities: The React frontend manages user interactions, including uploading outline sketches and entering text prompts. The Flask backend handles all processing tasks, including image refinement, interaction with AI models (Cycle GAN for sketch refinement, ControlNet with FLUX.1-dev for image generation), and performance metric evaluations.
Real-Time Features: WebSockets are integrated to allow users to monitor the status of image generation in real-time, enhancing interactivity and responsiveness.
Model Pipeline: Tracefy follows a multi-stage pipeline involving:
– ***Sketch Refinement:*** The input outline sketch is refined using a Cycle GAN-based model.
– ***Image Generation:*** The refined sketch and user-provided text prompt are input to ControlNet model (FLUX.1-dev-ControlNet-Union-Pro-2.0) guided by FLUX.1-dev for realistic face generation.

4.5. Assumptions and Dependencies

- Accurate User Inputs: The quality of the generated sketches heavily relies on the accuracy and completeness of the visual attributes provided by users.

- AI Model Hosting (Hugging Face): The system assumes the availability and reliability of AI model hosting on Hugging Face for inference. Downtime or API limits could impact sketch generation.
- Cloud Infrastructure Availability: The system is dependent on cloud infrastructure to handle large-scale operations and GPU-accelerated tasks.

4.6. Technical Stack

The Tracefy system is built using a carefully selected set of technologies spanning web development, machine learning, model evaluation, data storage, and deployment infrastructure. Each component of the technology stack is chosen to support the system's modular design, development efficiency, scalability, and performance.

4.6.1. Frontend Development

React.js

The frontend of Tracefy is developed using React, a modern JavaScript library for building responsive and component-based user interfaces. React enables a highly interactive user experience, allowing users to:

- Upload facial sketches.
- Enter and submit descriptive text prompts.
- View real-time image generation results.
- Navigate a seamless and reactive interface with stateful interactions.

The use of React ensures a scalable, maintainable, and performance-optimized frontend that supports real-time updates and asynchronous workflows.

4.6.2. Backend Development

Flask (Python)

The backend server is implemented using Flask, a lightweight and flexible Python web framework. Flask handles key backend responsibilities such as:

- Receiving and processing user inputs (sketches and text).
- Managing API endpoints for model inference requests.
- Orchestrating communication between the frontend, model pipeline, and database.
- Securing and validating incoming data.

Flask's simplicity and extensibility make it well-suited for integrating AI pipelines and web-based inference in a modular way.

4.6.3. Machine Learning & Deep Learning Frameworks

PyTorch

PyTorch is used extensively across the core modeling pipeline, including:

- Fine-tuning the FLUX.1-dev diffusion model.
- Integrating and deploying ControlNet and LoRA modules.
- Implementing image processing utilities and custom training routines.

PyTorch's dynamic computation graph and robust ecosystem make it ideal for experimentation, model customization, and integration with pretrained components.

TensorFlow

TensorFlow complements PyTorch in certain components where existing pre-trained models or libraries (e.g., some Cycle GAN implementations) are better supported. It is particularly useful in the sketch refinement stage, where stable TensorFlow-based Cycle GAN versions are leveraged.

4.6.4. Model Training Environment

Google Colab

.1. Model training, experimentation, and fine-tuning are conducted on Google Colab, which provides free and paid access to GPU and TPU resources. Colab is used for:

- LoRA fine-tuning of FLUX.1-dev and ControlNet models.
- Running Jupyter notebooks for data preprocessing and visualization.
- Rapid iteration and prototyping of training pipelines in a reproducible environment.

4.6.5. Database & Data Storage

MySQL

A MySQL database is used for managing structured data such as:

- User accounts and session data.
- Prompt history and generation metadata.
- Feedback and model evaluation logs.

MongoDB

MongoDB, a NoSQL database, is used in parallel to store unstructured or semi-structured data including:

- Sketches and generated image records.
- Annotation documents or custom prompt sets.
- JSON-based model configuration and tuning logs.

This hybrid database strategy combines the relational power of SQL with the flexibility of document-based storage to support both operational and research needs.

5. System Features and Requirements

5.1. Functional Requirements

5.1.1. REQ-1: User Registration and Authentication

The system shall allow users to create an account with a username, password, and email address.

Test:

- Attempt to create an account with valid details.
- Verify successful account creation.
- Attempt to create an account with missing or invalid details.
- Verify that account creation fails and appropriate error messages are displayed.

The system shall authenticate users using their username and password.

Test:

- Attempt to log in with valid credentials.
- Verify successful login.
- Attempt to log in with invalid credentials.
- Verify that access is denied and an appropriate error message is displayed.

The system shall allow users to reset their password via email verification.

Test:

- Initiate a password reset request.
- Verify that an email with reset instructions is received.
- Complete the password reset process.
- Verify that the user can log in with the new password.

5.1.2. REQ-2: User Input for Visual Attributes

The system shall provide a text input field where users can describe visual attributes such as age, gender, hair color, eye shape, facial structure, and distinguishing marks in natural language.

– **Test:**

(a) Access the text input interface.

(b) Enter detailed visual attribute descriptions in natural language (e.g., "young woman with long black hair and round glasses").

(c) Verify that the system processes the input and incorporates the attributes into the generated face image.

The system shall accept an initial outline sketch uploaded by the user and a descriptive text prompt to guide the image generation process.

– **Test:**

(a) Upload an outline sketch and enter a descriptive prompt.

(b) Verify that the system accepts and displays both inputs.

5.1.3. REQ-3: Sketch Generation

The system shall refine the input sketch using a CycleGAN-based model to produce a cleaner, more consistent sketch.

– **Test:**

(a) Upload a raw outline sketch.

(b) Verify that the system returns a refined sketch based on the original input.

The system shall generate a high-resolution facial image based on the refined sketch and user-provided description using a ControlNet model guided by FLUX.1-dev.

– **Test:**

(a) Complete the sketch refinement and submit the final prompt.

(b) Verify that a high-quality face image is generated.

5.1.4. Bias and Fairness Mitigation

The system shall aim to reduce demographic bias in generated outputs by incorporating diverse training data and fair prompt representation.

– Test:

(a) Input prompts representing various ethnicities, ages, and genders.

(b) Verify that the outputs show balanced and fair visual representations.

The system shall support ongoing evaluation of fairness using performance metrics across different demographic groups (e.g., FID, LPIPS, Canny similarity).

– Test:

(a) Generate outputs for diverse user groups.

(b) Use evaluation metrics to assess and compare fairness and quality.

5.1.5. User Interface

The system shall offer a user-friendly web interface built in React, allowing users to upload sketches, input prompts, and view generated outputs.

– Test:

(a) Navigate the frontend interface.

(b) Verify intuitive interactions and visual clarity of results.

The system shall provide a secure dashboard for authorized users (e.g., law enforcement) to manage generated images and related input data.

– Test:

(a) Log in as a law enforcement user.

(b) View, search, and manage stored results via the dashboard.

5.1.6. Data Management

The system shall securely store all user inputs (sketches and prompts) and generated images in a backend database with appropriate access controls.

– Test:

(a) Generate a new image from user input.

(b) Verify that the input and result are stored and secured.

The system shall allow users to retrieve previously submitted sketches, prompts, and corresponding outputs.

– Test:

(a) Log in and access the history feature.

(b) Verify that prior inputs and generated images are correctly retrieved.

5.2. Non-Functional Requirements

5.2.1. Performance

- The system shall generate a response within a specified timeframe for standard queries related to sketch generation.
  - Test:
  - (a) Measure the time taken from input submission to output generation for various queries.
  - (b) Verify that the response time is consistently within the specified limit across multiple tests.
- The system shall handle a minimum number of concurrent users without any noticeable delays in response time.

- Test:
- (a) Simulate the specified number of concurrent users performing sketch generation tasks.
- (b) Monitor and verify that all users receive responses without significant delays.

5.2.2. Reliability

- The system shall maintain a high level of uptime to ensure availability for emergency situations.
  - Test:
  - (a) Monitor system uptime over a specified period.
  - (b) Verify that downtime does not exceed the allowed threshold.
- The system shall implement automatic failover mechanisms to recover from server failures within a specified timeframe.
  - Test:
  - (a) Simulate a server failure scenario.
  - (b) Measure the time taken for the system to recover and resume operations.

5.2.3. Security

- The system shall employ encryption for data at rest and secure protocols for data in transit to ensure user data protection.
  - Test:
  - (a) Verify that data stored in the database is encrypted.
  - (b) Monitor network traffic to ensure secure protocols are used for all data transmissions.
- The system shall support role-based access control (RBAC) to restrict access to sensitive information based on user roles.
  - Test:
  - (a) Test user roles with varying permissions to access sensitive data.
  - (b) Verify that users cannot access information outside their assigned roles.

5.2.4. Compliance

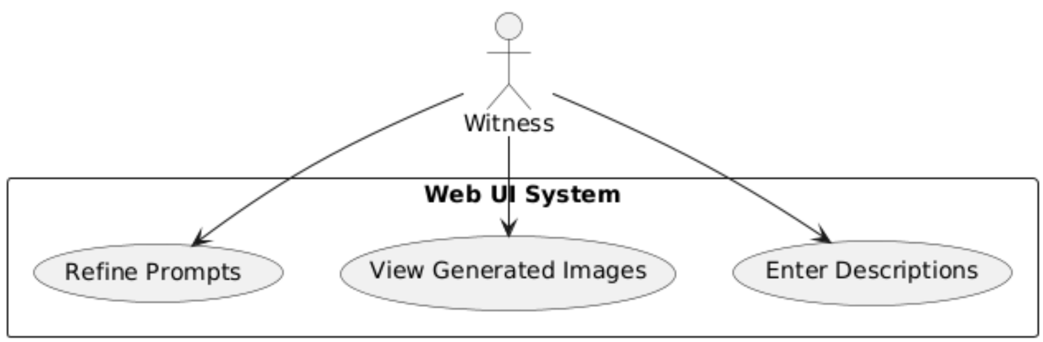
- The system shall comply with relevant regulations concerning the handling of sensitive personal information in law enforcement contexts.
  - Test:
  - (a) Conduct a compliance audit to ensure all data handling practices meet regulatory standards.
  - (b) Verify that appropriate user permissions and access logs are maintained.
- The system shall adhere to cybersecurity frameworks to ensure robust security practices.
  - Test:
  - (a) Assess the security controls in place against the framework guidelines.

5.2.5. Ethical Considerations

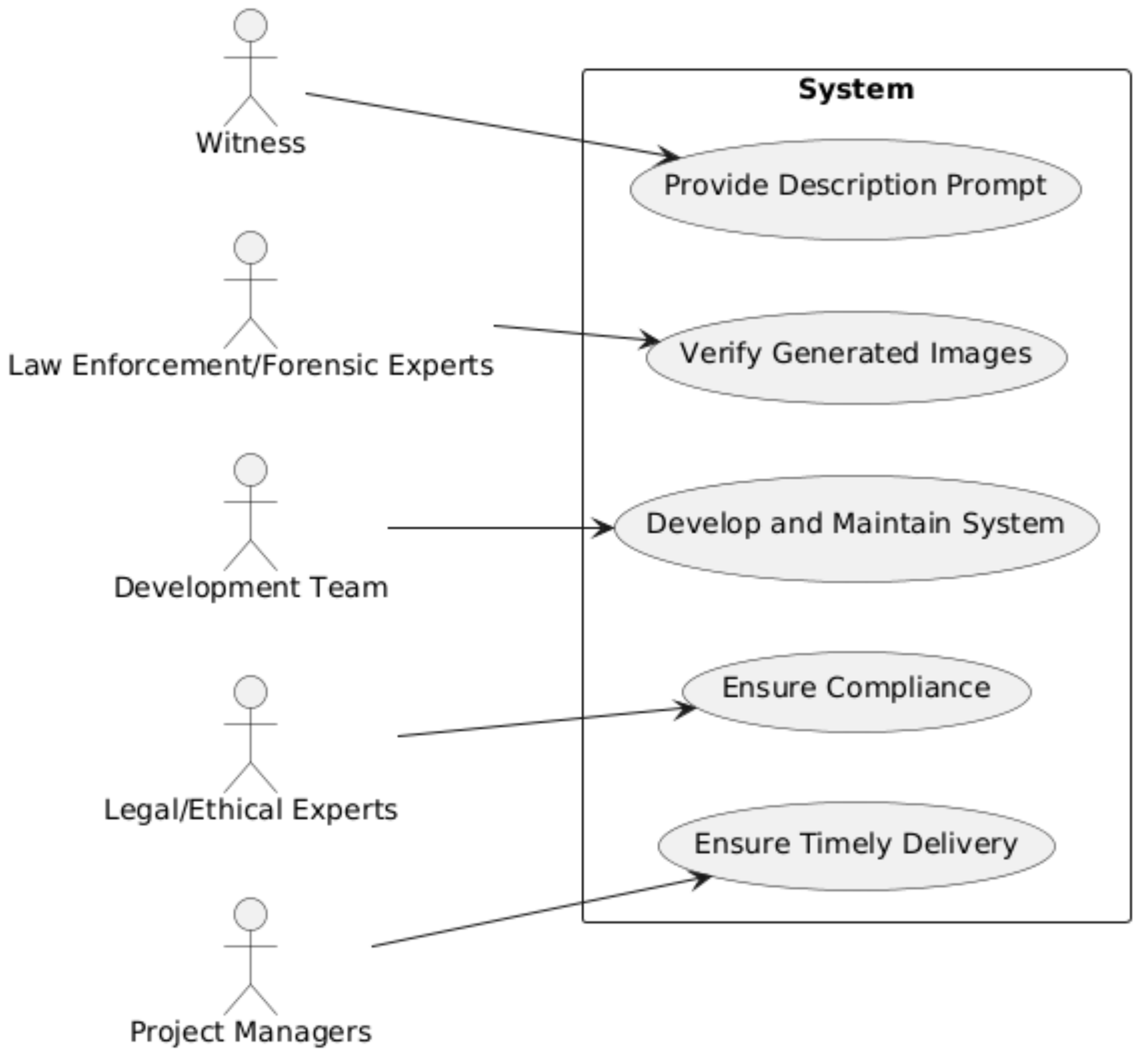
- The system shall incorporate fairness metrics in the algorithm to assess and minimize biases in generated sketches based on demographic data.
  - Test:
  - (a) Evaluate the algorithm for fairness metrics during the sketch generation process.
  - (b) Analyze generated sketches for demographic representation and potential biases.
- The system shall provide transparency in the sketch generation process, including disclosures on how data is used and how fairness is ensured.
  - Test:
  - (a) Review user documentation for clarity on data usage and fairness measures.
  - (b) Confirm that users can access information regarding the generation process and fairness checks.

5.3. Use Case Diagrams

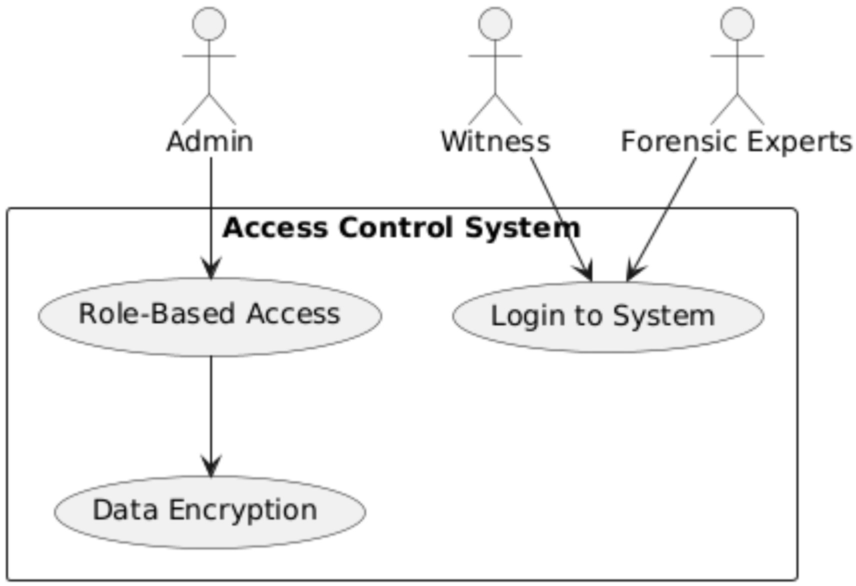
5.3.1. User Interface (UI) Requirements



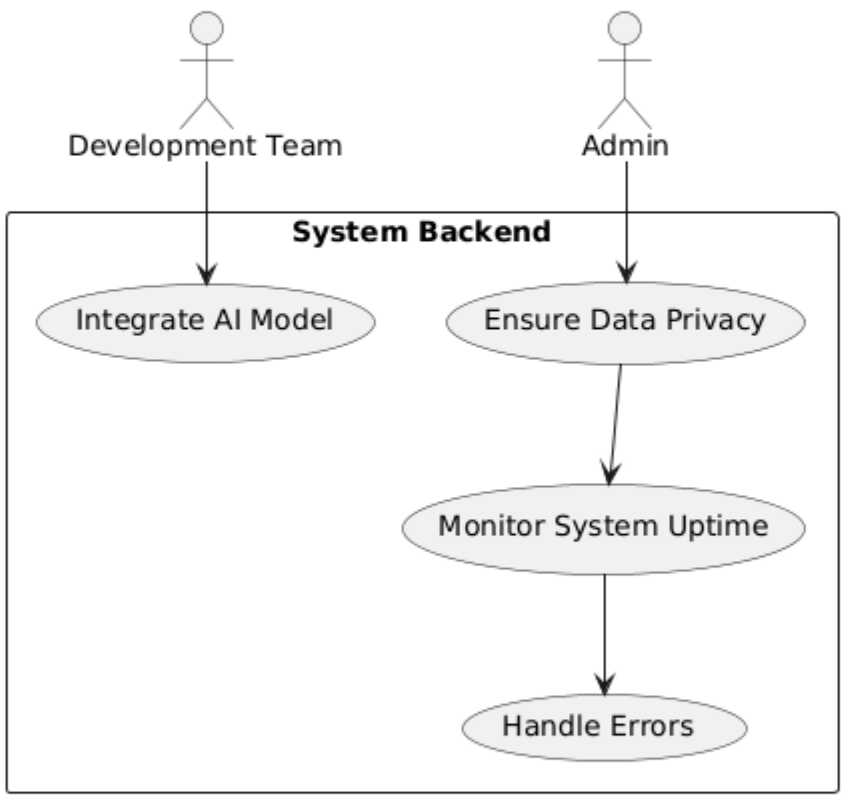
5.3.2. Stakeholder Identification



5.3.3. Authentication and Access Control

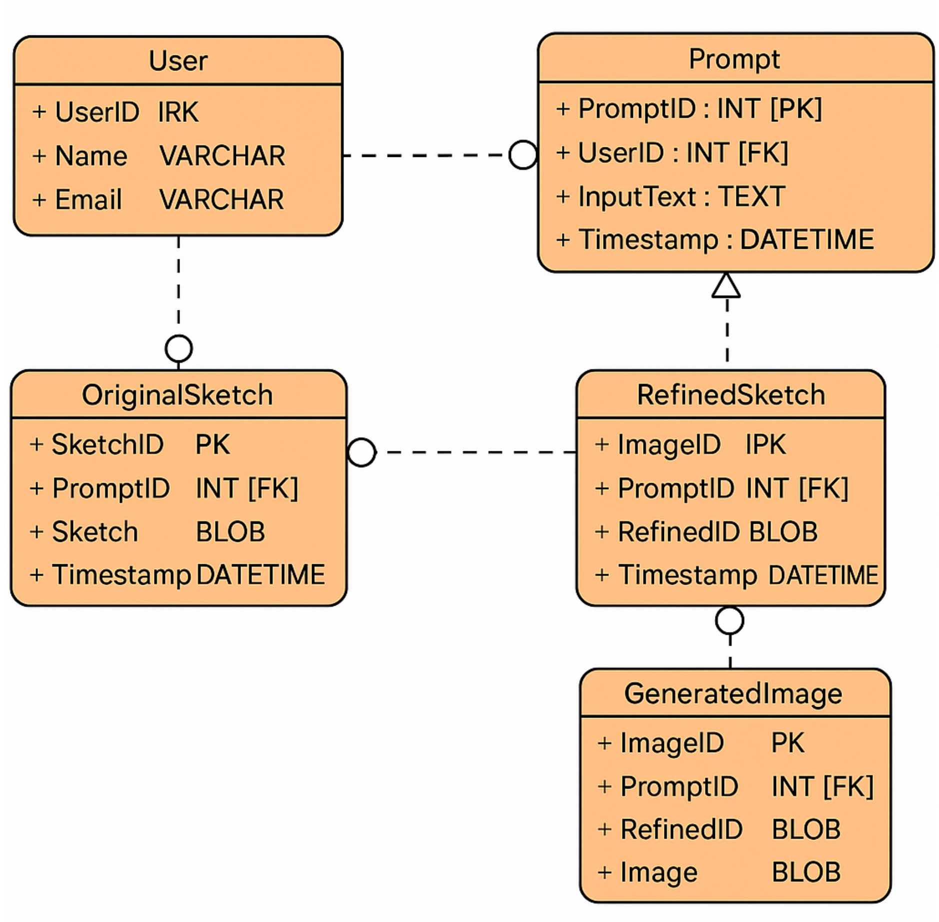


5.3.4. Design and Security

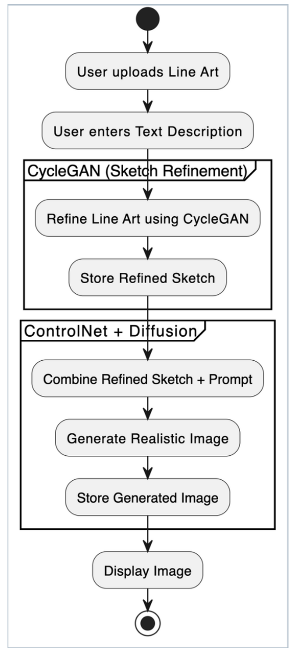


6. Design Models

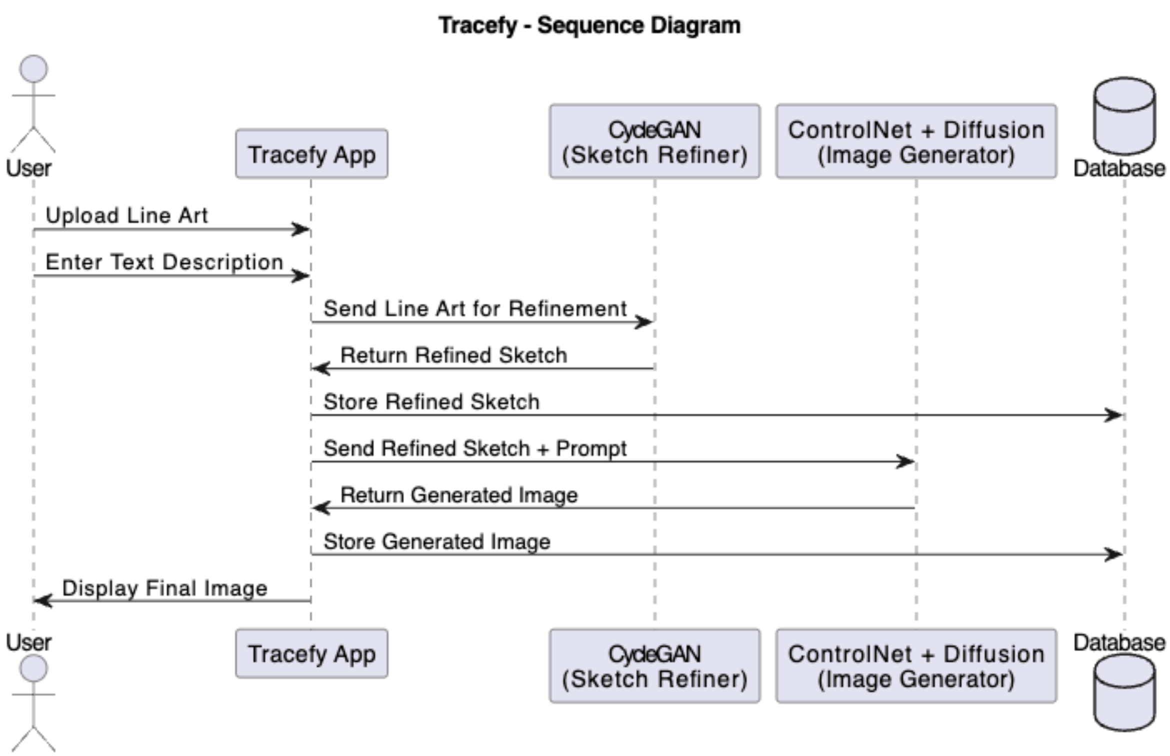
6.1. ERD and Schema



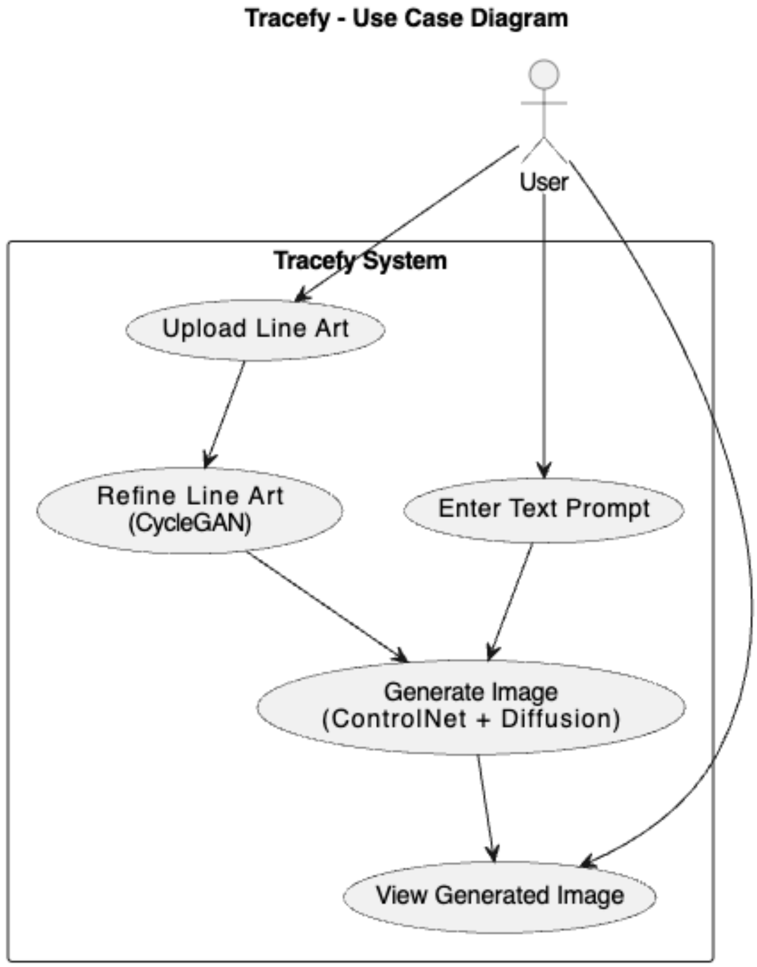
6.2. Activity Diagram



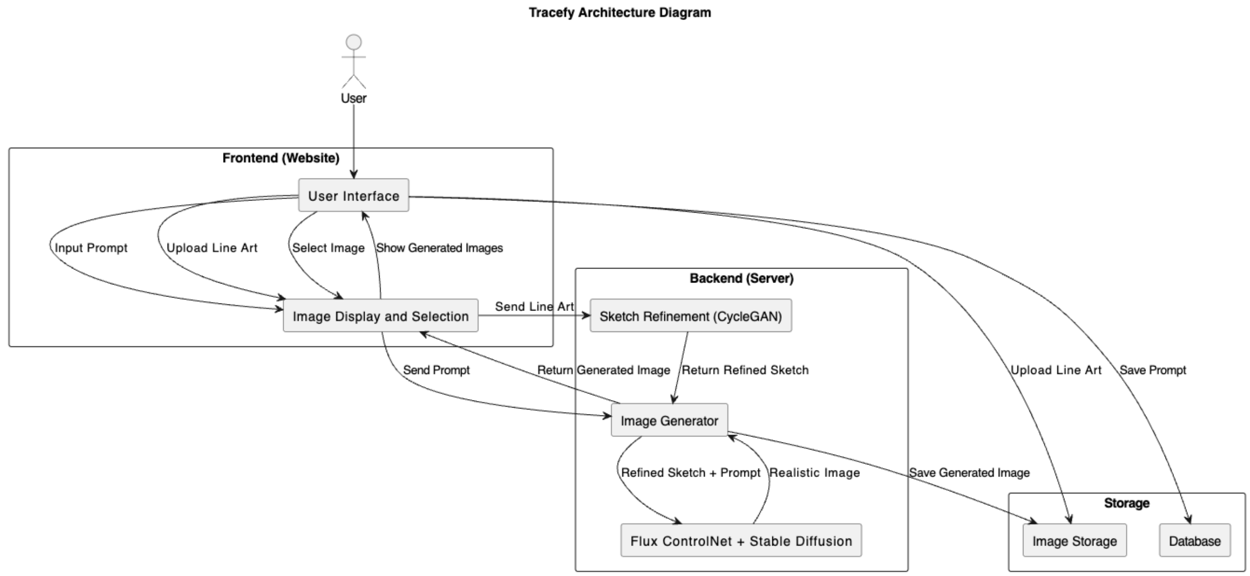
6.3. Sequence Diagram



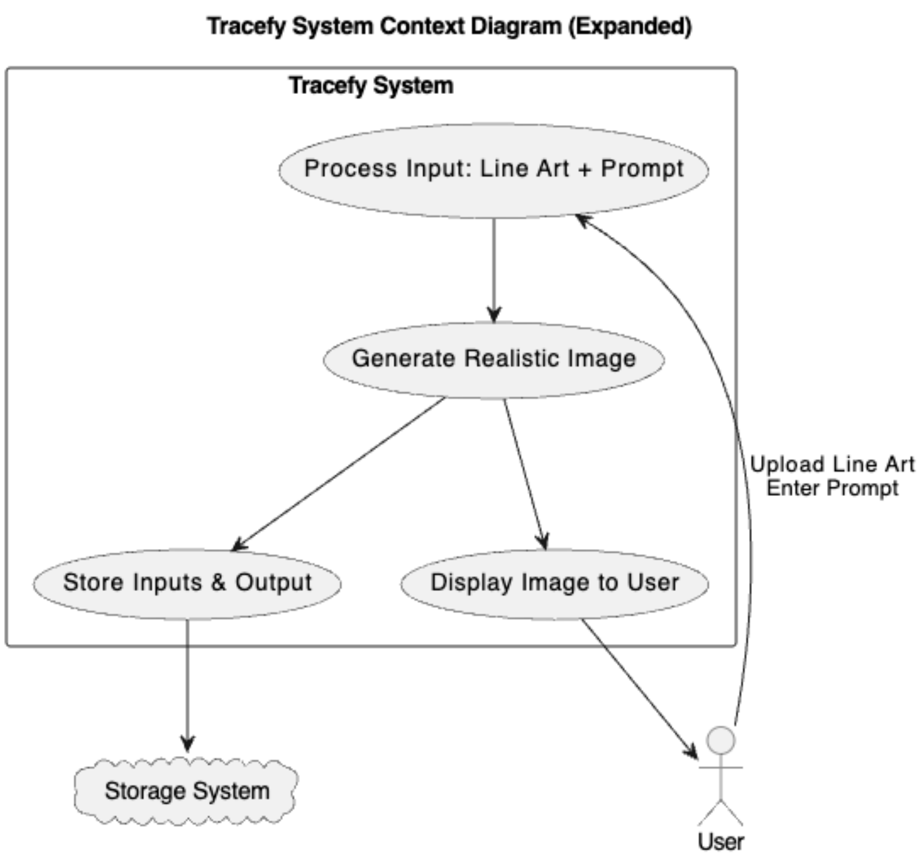
6.4. Use Case Diagram



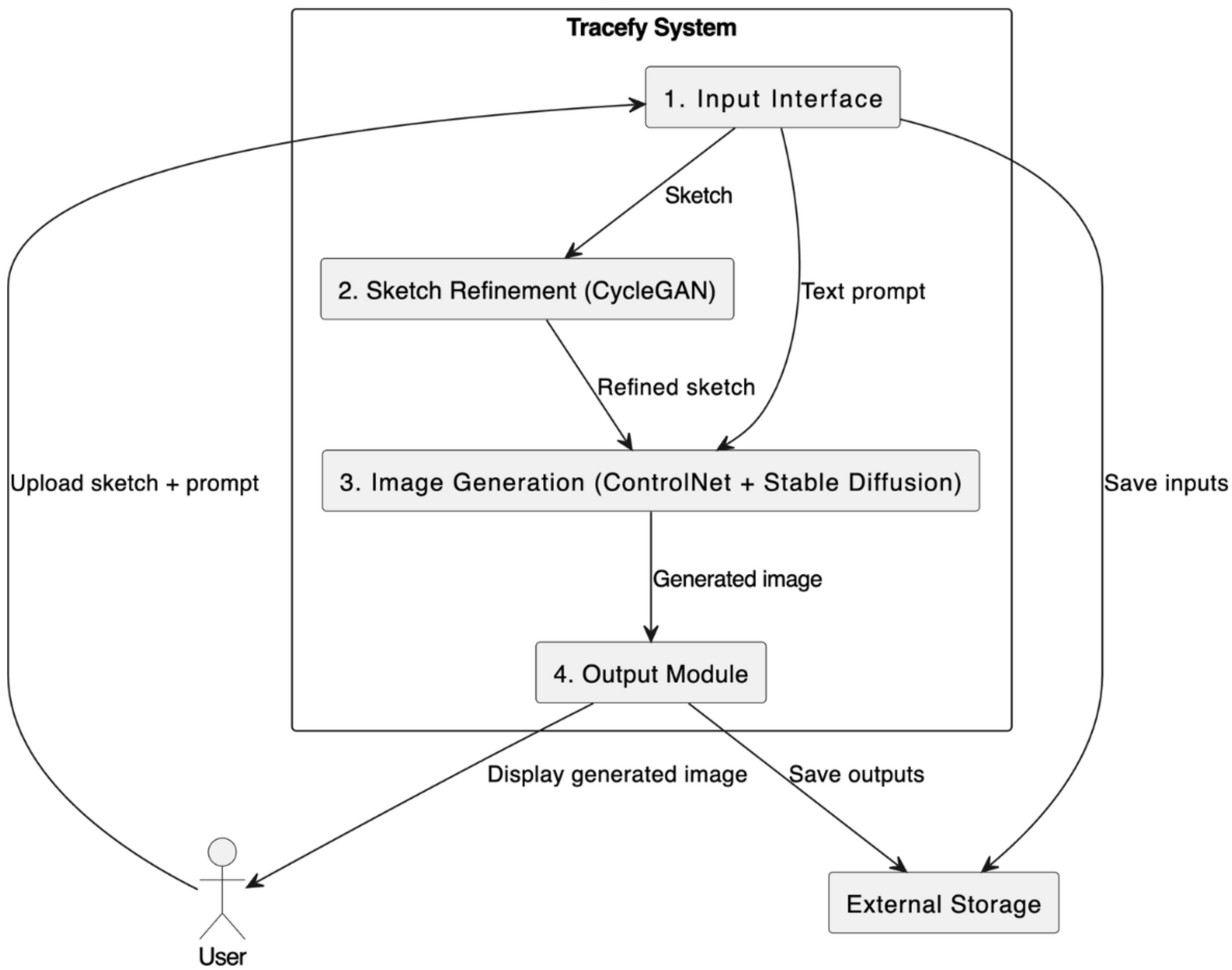
6.5. Architectural Diagram



6.6. Context Diagram



6.7. DFD Level 1 Diagram



7. Data Design

7.1. Data Collection

The data used for training the sketch refinement model in Tracefy was sourced from the CUHK Face Sketch (CUFS) dataset, which contains paired images of real face photographs and artist-drawn sketches. To adapt the dataset for use with a Cycle GAN model, the dataset was reorganized into two distinct domains:

Domain A – Line Art Sketches: These represent the "raw" or outline sketches that simulate user-drawn input. To create this domain:

The original CUFS photographic images were first processed using the Sobel filter to enhance edge features. Then, the filtered images were passed through a SimpleSketch pretrained model, which outputs simplified line-art representations. These line-art sketches simulate the type of input a user might upload into Tracefy.

Domain B – Refined Sketches: These are cleaner, more complete sketches generated by artists and present in the CUFS dataset. They serve as the target domain during Cycle GAN training, representing how a high-quality sketch should appear after refinement.

By establishing these two domains, the Cycle GAN model was trained to translate noisy, user-style input sketches (Domain A) into high-quality, clean sketches (Domain B) without requiring paired examples for every training step.

7.2. Data Preprocessing

To ensure consistency and model performance, several preprocessing steps were applied: Resizing and Cropping: All images were resized and center-cropped to a uniform resolution (e.g., 256×256 pixels), which is compatible with the Cycle GAN input requirements.

Grayscale Conversion: Both domains were standardized to grayscale to eliminate irrelevant color information, focusing the model on edge and structure representation.

Normalization: Pixel values were normalized to the range [-1, 1] to match the expectations of the neural network layers and accelerate convergence.

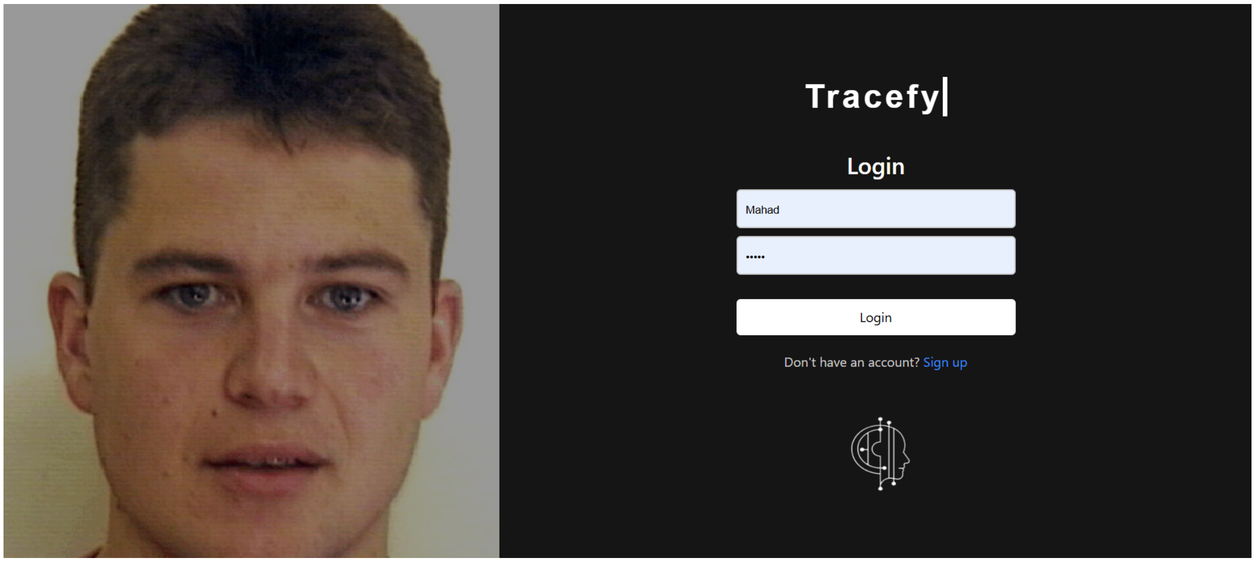
Data Augmentation: Techniques such as horizontal flipping and slight rotation were applied during training to improve generalization and reduce overfitting.

Batch Formatting: Images were grouped into mini-batches and shuffled for efficient GPU training using PyTorch DataLoaders.

This carefully structured data design ensured that the CycleGAN model could learn an effective transformation between rough input sketches and refined outputs, which is a core step in the Tracefy pipeline.

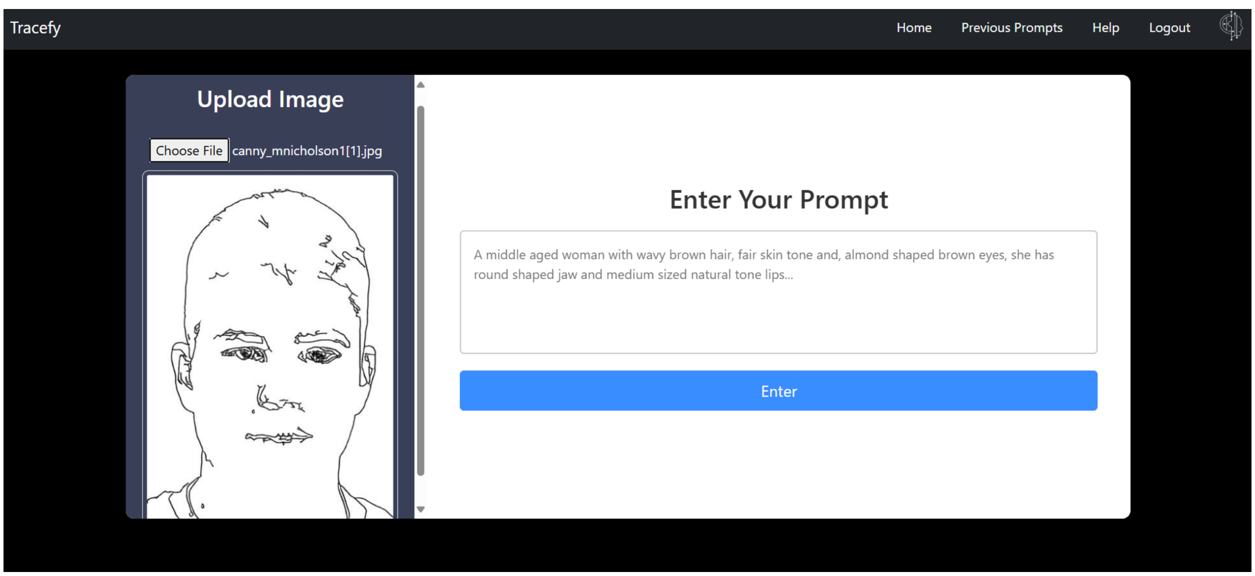
8. User Interfaces

Tracefy's input interface begins with a secure login page, ensuring that only authorised individuals, typically law enforcement authorities, have access to the 38 system. This authentication stage is critical since Tracefy is built as a B2G (Business-to-Government) solution that ensures privacy, data protection, and control over sensitive criminal investigation operations.



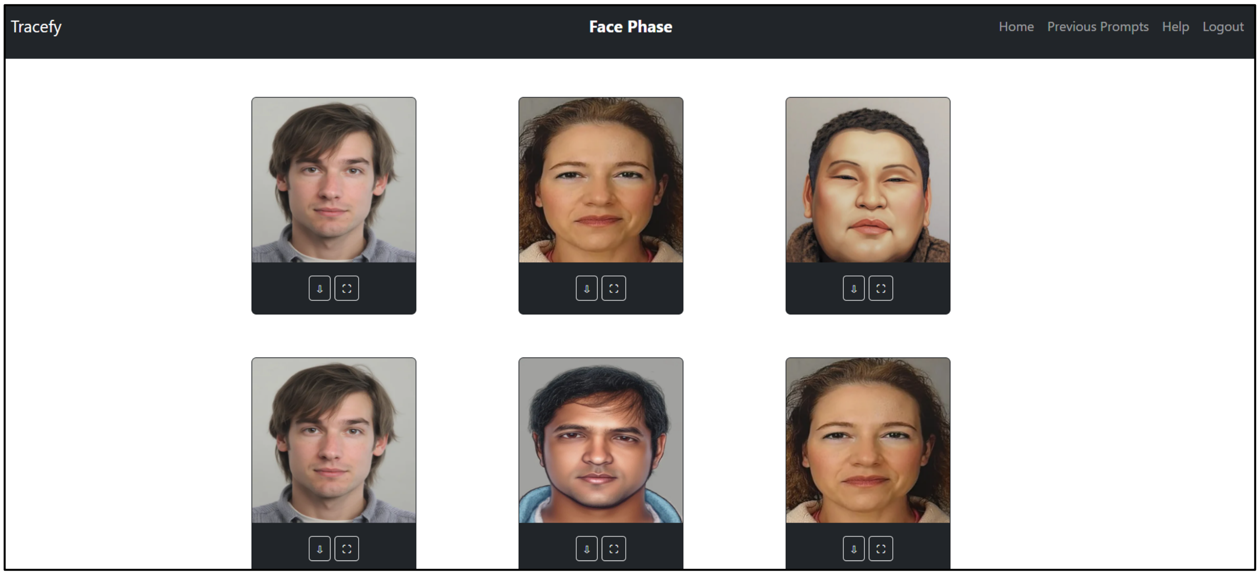
Login Page

After successfully logging in, participants are sent to a prompt screen where they can enter structured descriptions of suspects based on eyewitness testimony. This description serves as the primary input to the AI model. Once submitted, the prompt is processed and routed to the picture generating module.



Main Dashboard

The AI model then creates a preliminary sketch based on the input. If the sketch matches the user's expectations, it moves on to the refinement step, when it is coloured and transformed into a realistic image. If not, the system allows for iterative sketch generation using the original prompt until the user is satisfied with the outcome.



Output Image Page

9. Features of the Final Product

The final implementation of Tracefy delivers a robust and interactive platform that integrates user-friendly design with advanced AI-driven image synthesis. The core features of the system include:

9.1. User Authentication and Login:

Users are required to register and log in to access the system. Authentication mechanisms ensure secure access and protect user-generated data.

9.2. Sketch Upload Interface:

Upon logging in, users can upload an initial line-art sketch representing the suspect. This sketch acts as the base visual input to guide the generation process.

9.3. Text-Based Suspect Description:

Users can enter a natural language prompt describing key visual attributes of the suspect—such as age, gender, hairstyle, facial features, and distinguishing marks. This description complements the uploaded sketch and guides the image synthesis pipeline.

9.4. Sketch Refinement View:

The uploaded line-art sketch is automatically refined using a CycleGAN model to produce a cleaner, more structured version. The refined sketch is displayed in the interface for user review.

9.5. Facial Image Generation:

Once the refined sketch and prompt are confirmed, they are sent to the backend server, where the FLUX.1-dev-ControlNet model generates a realistic facial image. This image reflects both the structural features of the sketch and the descriptive text input.

9.6. Output Viewing and Download:

The generated facial image is presented to the user within the interface and can be downloaded for external use or further analysis.

9.7. Prompt Iteration and Refinement Loop:

To ensure the generated image closely resembles the suspect as witnessed, users can refine and resubmit the prompt up to 6 times per session. Each iteration updates the image generation while retaining the original sketch context, allowing for progressive refinement.

This feature set allows Tracefy to serve as a practical and user-friendly tool for law enforcement and forensic applications, transforming rough witness input into high-fidelity, realistic face images.

10. Development Workflow and Management Tools

10.1. Project Management

Effective project management and version control were essential to the successful development of Tracefy. For project management, Trello was utilized to organize tasks, track progress, and collaborate efficiently among team members. Trello's card-based interface allowed the team to break down the project into manageable tasks across different phases such as design, development, testing, and deployment. Features like task assignment, due dates, checklists, and progress tracking via boards and lists helped maintain clarity and accountability throughout the development cycle.

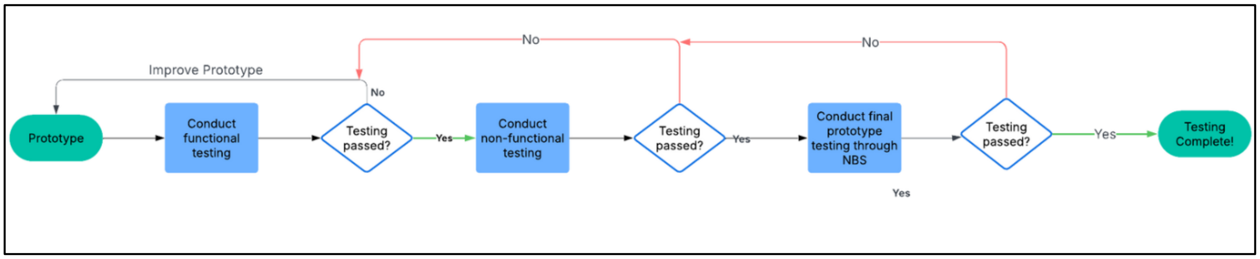
10.2. Version Control

For version control, GitHub was employed to manage and track changes in the codebase. GitHub facilitated collaborative development by allowing multiple contributors to work on different branches, submit pull requests, and perform code reviews. Key features like commit history, issue tracking, and project wikis supported robust documentation and debugging. GitHub Actions also enabled automation for testing and deployment workflows. Together, Trello and GitHub provided a streamlined and transparent workflow, ensuring smooth coordination and consistent progress in the Tracefy project.

11. Testing

The testing phase for Tracefy was carefully designed to ensure that the system performs reliably, generates accurate outputs, and provides a positive user experience across both technical and practical dimensions. The primary objectives of testing included validating functionality, evaluating sketch quality, identifying bias, and collecting user feedback. The system was assessed for usability by both law enforcement personnel and general users through controlled trials. Additionally, testing efforts were supported by NUST Business School, which helped simulate real-world usage and facilitated the collection of structured feedback.

Tracefy's testing was divided into different levels, ranging from unit testing of individual components to full pipeline testing of the entire user journey. Each level focused on different aspects, such as the correctness of feature mappings, the flow between interface and backend, and the performance of the system in live usage scenarios.



Testing Plan

11.1. Prototype Testing

Initial prototype testing was conducted in a controlled environment with student volunteers acting as test users. Participants were asked to describe fictional individuals using Tracefy's structured prompt system. These descriptions were then processed by the system to generate AI-based facial sketches. The goal was to mimic the role of a witness reporting a suspect and to observe how effectively the system could interpret the input into visual form.

Users evaluated the output based on resemblance to their intended description, visual quality, and accuracy of individual attributes. Most participants found the system intuitive and appreciated the guided input interface, which reduced ambiguity in descriptions. Sketch generation time ranged from 2 to 5 minutes depending on model load, and most sketches matched user expectations when inputs were specific and descriptive. Some users, however, requested more detailed options for facial features like hair texture and skin tone, especially to better reflect South Asian diversity.

11.2. Unit Testing

At the most granular level, unit testing was conducted to verify whether individual modules of the system functioned correctly. The focus was on ensuring that each facial attribute selected—such as eye shape, nose type, or hairstyle—produced the corresponding visual effect in the final sketch. Input validation was also tested by submitting incomplete or incorrect descriptions to check whether the system appropriately handled errors or guided the user to correct them. This helped confirm the robustness of the prompt interface and the underlying logic that maps descriptors to visual features.

11.3. Integration Testing

Integration testing focused on how well different components of the system worked together. This included ensuring that the user input collected through the frontend interface was accurately transmitted to the backend sketch generation engine. Test sessions confirmed that the descriptions entered by users were processed correctly and that the generated image was properly rendered and displayed within the same session. Furthermore, transitions between key stages—such as from description to image, and from image to download or submission—were smooth and without unexpected behavior. These tests helped validate that Tracefy operates as a cohesive system, not just a set of disjointed features.

11.4. Full Pipeline Testing

Full pipeline testing evaluated the system end-to-end, covering the entire user experience from the initial prompt to the final output. This included observing how real users navigated the interface, submitted descriptions, reviewed sketches, and interpreted results. This phase also involved collecting qualitative feedback on whether the tool felt practical and usable in a real-world investigative setting. A special focus was placed on feedback from individuals simulating law enforcement officers, who emphasized clarity of the interface and realism in generated outputs. The system performed reliably, though some minor adjustments were identified for improving the specificity of visual traits for certain regional facial features.

11.5. Non-Functional Testing

Beyond functional requirements, several non-functional aspects were also assessed. Performance testing involved measuring sketch generation time across different sessions to ensure it remained consistent. Usability testing was conducted by observing how users interacted with the interface—whether they could easily navigate the dropdown menus, understand the prompts, and complete the sketch generation process independently. Error handling was tested by entering incomplete or invalid data to see how gracefully the system responded. In all cases, users received clear feedback or prompts to complete their input, ensuring no system crashes or dead ends. Accessibility was reviewed informally, focusing on text visibility, color contrast, and intuitive layout. While not formally tested with assistive technologies, the system was found to be visually clear and navigable.

While Tracefy is still in its early stages, this testing phase laid the foundation for identifying strengths and areas for refinement. Scalability and deployment readiness will be addressed in future iterations once the model and dataset reach higher maturity.

Testing Type	**Focus Area**	**Outcome**
Prototype Testing	Prompt-to-sketch accuracy and realism	Users found it usable and visually accurate
Unit Testing	Input validation and feature mapping	Components responded correctly to valid/invalid input
Integration Testing	Communication between interface and backend	Inputs and outputs worked together without errors
Full Pipeline Testing	End-to-end functionality from user input to output	System worked reliably; interface was intuitive

Performance Testing	Sketch generation time and consistency	Average time was 2–5 minutes
Usability Testing	Ease of navigation, user understanding, and comfort	Generally intuitive; requests for more facial details
Error Handling	Behavior during missing or invalid input	Clear prompts guided users; no crashes observed
Accessibility (basic)	Readability, contrast, layout	Clear layout and good contrast

Table 2: Description of Different Testing Plans

12. Model Architecture – Cycle GAN

12.1. Why Cycle GAN?

In the context of forensic face synthesis, refining a user-drawn line art into a realistic sketch requires a model that can handle unpaired, imprecise, and stylistically diverse inputs. Several image-to-image translation models exist—namely Pix2Pix, StarGAN, and SPADE—but they fall short in addressing the specific requirements of Tracefy. After careful evaluation, Cycle GAN emerges as the most suitable model for this task.

12.2. Comparison with other Img2Img Models

12.2.1. Pix2Pix

Pix2Pix is a supervised image-to-image translation model that requires paired datasets (i.e., exact sketch-photo pairs). This is a major limitation in forensic scenarios where such pairs are rarely available and sketches may not be perfectly aligned with the real facial features.
Limitation: Requires aligned paired data, which is unrealistic for forensic or user-drawn sketches.
Result: Poor performance on noisy or misaligned sketches.

12.2.2. Star GAN

Star GAN supports multi-domain image translation using a single generator, which is effective in some cases but requires domain labels (e.g., gender, age, emotion). Additionally, it struggles to preserve fine facial detail such as eye and mouth structure, which are critical in sketch-to-face translation.
Limitation: Requires labeled domains and often produces less detailed facial outputs.
Result: Not robust enough for forensic applications where fine detail is crucial.

12.2.3. SPADE (Spatially-Adaptive DE-normalization)

SPADE is designed for high-quality image synthesis from semantic segmentation masks. While powerful for photorealistic rendering, it fails on raw, unstructured sketches and lacks the flexibility to produce stylized or sketch-like outputs, which are often needed for intermediate forensic visualization.
Limitation: Incompatible with raw sketches and lacks sketch-style flexibility.
Result: Poor refinement of hand-drawn or forensic-style line art.

Model	Paired Data Required	Works with Raw Sketches	Detail Preservation	Requires Domain Labels	Suitable for Tracefy?	Limitations
Pix2Pix	Yes	No	Moderate	No	No	Requires aligned paired data; performs poorly on unaligned or user-drawn sketches
StarGAN	No	No	Weak	Yes	No	Needs labeled domains; fails to retain fine facial details
SPADE	No	No	High (on segmentation masks)	No	No	Fails on unstructured sketches; not flexible for sketch-style outputs
CycleGAN	No	Yes	Strong	No	Yes	Slightly less photorealistic than supervised methods, but more robust and flexible

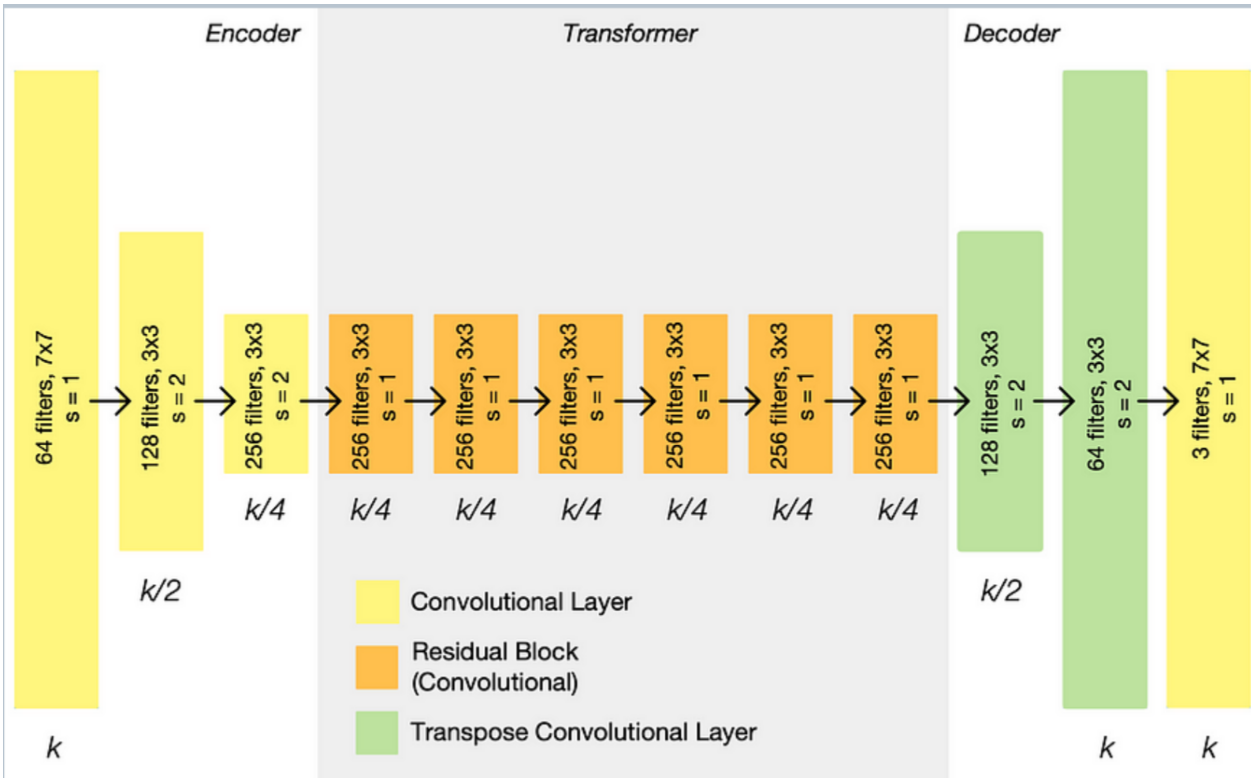
Table 3: Comparison of Image to Image Models

12.3. Img2Img Translation using Cycle GAN

To refine the input line art (face outline) into a more detailed sketch, Tracefy utilizes an image-to-image translation technique based on the Cycle GAN architecture, originally introduced by Jun-Yan Zhu et al. in 2017. Unlike supervised models that require paired input-output images, CycleGAN enables unpaired image translation, making it suitable for scenarios where paired training data is unavailable.

12.4. Architecture Overview

12.4.1. Generator Architecture: Encoder–Transformer–Decoder



Each generator in Cycle GAN follows a fully convolutional design comprising three main components:

Encoder

A series of convolutional layers with stride 2 to progressively reduce spatial resolution and extract deep features. Each layer is followed by Instance Normalization and ReLU activation.

Transformer (Residual Blocks)

A sequence of 6 or 9 residual blocks, each containing:

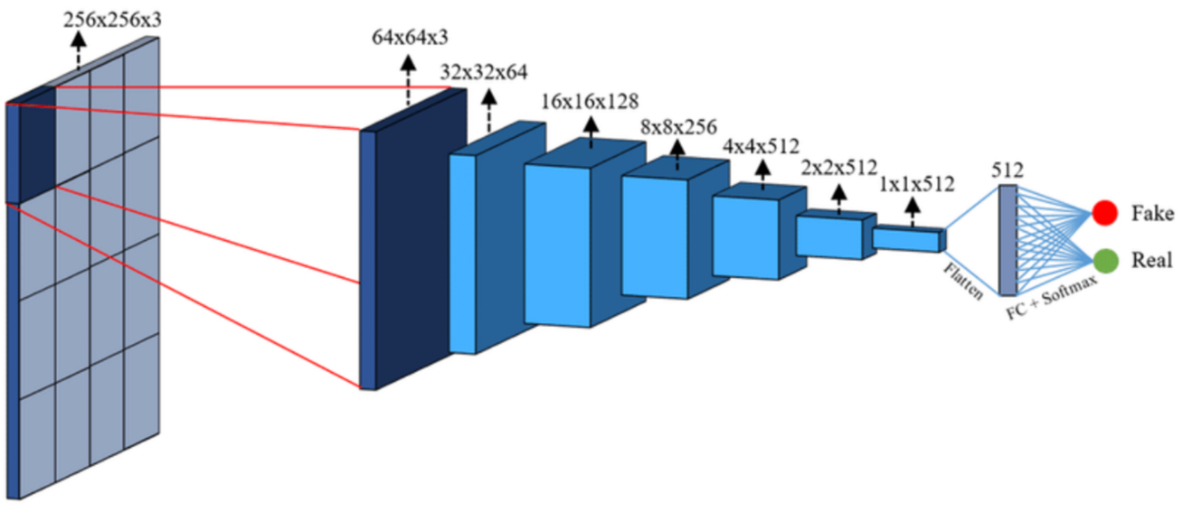
Conv → InstanceNorm → ReLU → Conv → InstanceNorm

A skip connection that adds the input of the block back to the output. These blocks allow the network to learn complex transformations while preserving structural features.

Decoder

Transposed convolutions (or up sampling followed by convolution) are used to up sample the feature maps back to the original image resolution. The final layer uses a Tanh activation function to constrain the output pixel values in the range [-1, 1].

12.4.2. Discriminator Architecture: Patch GAN



Each discriminator adopts the Patch GAN architecture, which evaluates the realism of local patches (e.g., 70×70 regions) rather than the entire image. This design encourages the network to generate high-frequency details and textures. Composed of several convolutional layers with:

- A kernel size of 4 and stride of 2 for down sampling.
- LeakyReLU activations.
- Instance Normalization (except in the first layer).

The final output is a 2D map where each value represents the probability that a corresponding patch in the image is real.

12.5. Model Training and Evaluation

To enhance the quality of user-provided line art sketches within the Tracefy pipeline, a CycleGAN model was trained for unpaired image-to-image translation. The objective was to learn mappings between Domain X (line art) and Domain Y (refined sketches) using unpaired datasets, allowing for more flexibility and applicability in forensic scenarios where paired data is scarce or unavailable.

Parameter	Value
Number of Epochs	50
Batch Size	1
Optimizer	Adam
Learning Rate	0.0002
Learning Rate Decay	Linear decay after half the total epochs
Early Stopping	Enabled to prevent overfitting
Loss Functions	Adversarial Loss, Cycle Consistency Loss
Input Data	Unpaired line art and refined sketch images

A batch size of 1 was selected to accommodate the high-resolution and detailed nature of sketch data. Early stopping was applied based on validation performance to prevent overfitting and improve generalization.

12.5.1. Adversarial Loss:

Encourages each generator to produce outputs that are indistinguishable from real samples in the target domain. Each generator-discriminator pair competes in a minimax game to improve realism in the generated images.

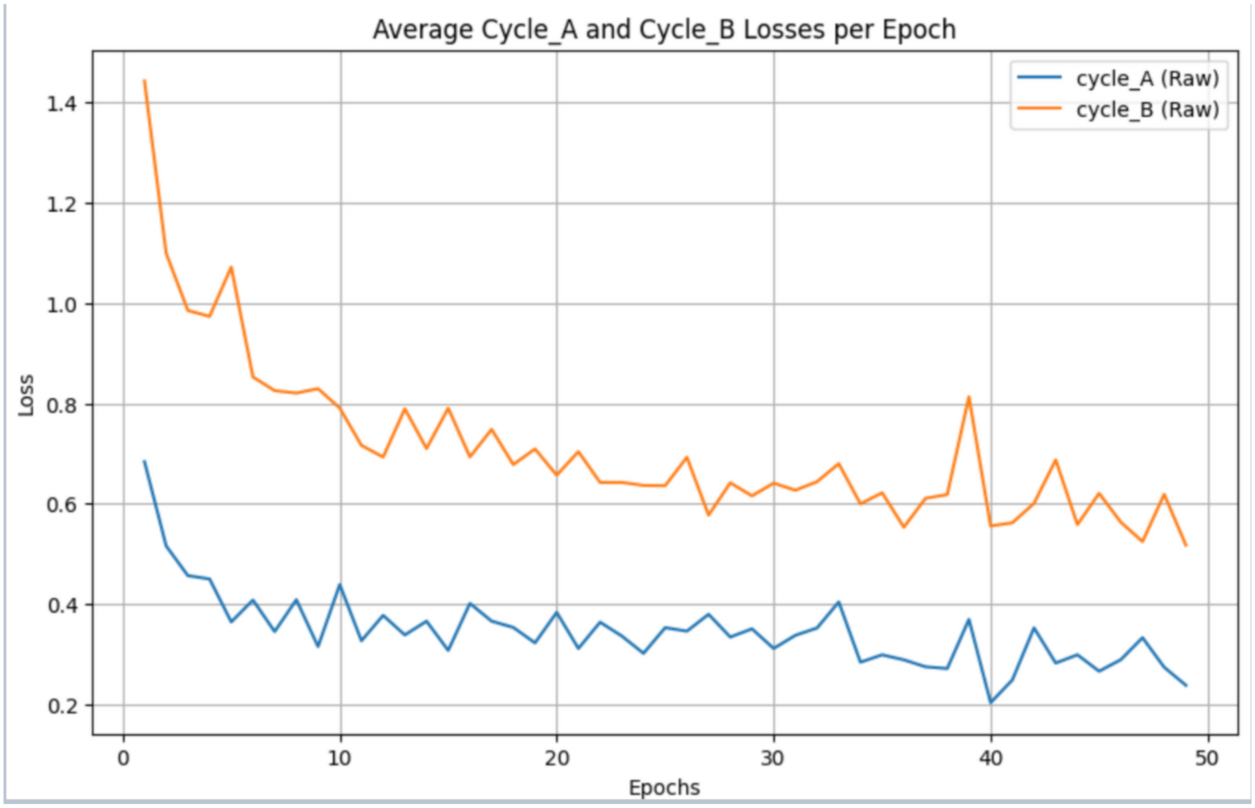
12.5.2. Cycle Consistency Loss:

Ensures that an image translated to the other domain and then back to the original domain remains consistent with its source. This loss is critical for maintaining structural fidelity in the absence of paired training data. The formula for cycle consistency loss is defined as follows:

$$Lcyc(G,F)=Ex\sim X[\|F(G(x))-x\|1]+Ey\sim Y[\|G(F(y))-y\|1]$$

12.5.3. Training Behavior

Throughout training, the model demonstrated a progressive improvement in its ability to reconstruct the original image after performing a cycle of forward and backward translation. The cycle consistency loss steadily decreased, indicating better preservation of key facial features such as eyes, nose, and mouth. Additionally, the adversarial loss guided the generators toward producing increasingly realistic and stylistically accurate sketches.



12.5.4. Results during the Training Process

Key Aspects from the graph above

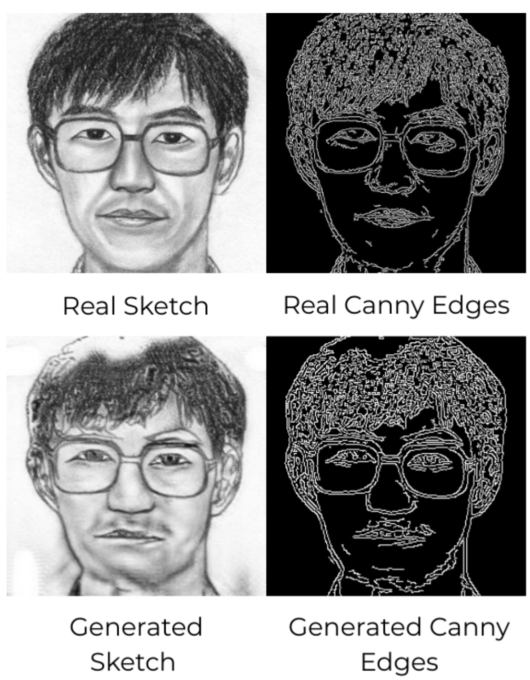
- Loss Reduction: Both cycle_A and cycle_B losses decrease over epochs, indicating successful learning.
- Asymmetry: cycle_B remains consistently higher — suggesting that going from line art to sketch is more challenging than the reverse.
- Training Stability: There's some fluctuation (esp. around epoch 40), but overall the losses remain on a downward trend.
-

The diagram below visually compares the results of the Cycle GAN model during the training process showing the improvement in refinement of sketch as the model approaches the final epochs.

12.5.5. Cycle GAN Results and Evaluation

The following results are obtained from the model which are evaluated using the Canny Edge Detection It evaluates how well the edges in the generated sketch align with those in the ground truth. Unlike pixel-wise metrics (e.g., MSE, PSNR), edge similarity focuses on structural alignment rather than exact pixel values.

Cycle GAN Results



Cycle GAN Evaluation using Canny Edge Detection

13. Model Architecture – Flux-Control Net

FLUX with ControlNet is particularly well-suited for Tracefy because it effectively combines text-guided generation with structure-aware control, ensuring that both the semantic intent from the prompt and the precise shape information from the sketch are faithfully preserved in the final image.

13.1. Why Flux Control Net?

1. Dual Conditioning with Precision

- Tracefy's goal is to convert a user-provided face sketch and descriptive text into a realistic facial image. This requires maintaining fidelity to both the outline/structure (from the sketch) and the attributes or style (from the text prompt).
- The text prompt is encoded and integrated via cross-attention into the FLUX U-Net, guiding the model to capture the intended identity, style, and features.
- The sketch or edge map is processed by ControlNet, which extracts structural features and injects them as residuals into multiple layers of the U-Net.
- This dual pathway ensures that the generation respects both the content (what the user wants) and the form (how the face should be shaped).

2. Layer-Wise Guidance from ControlNet

- ControlNet doesn't just guide the model at the input level—it injects learned residuals into the U-Net at multiple layers. This fine-grained structural conditioning helps maintain:
    - Facial geometry
    - Pose and proportions
    - Boundary accuracy (e.g., hairline, jaw, eye placement)
- This is crucial for Tracefy, where user-drawn outlines must directly correspond to the output image.

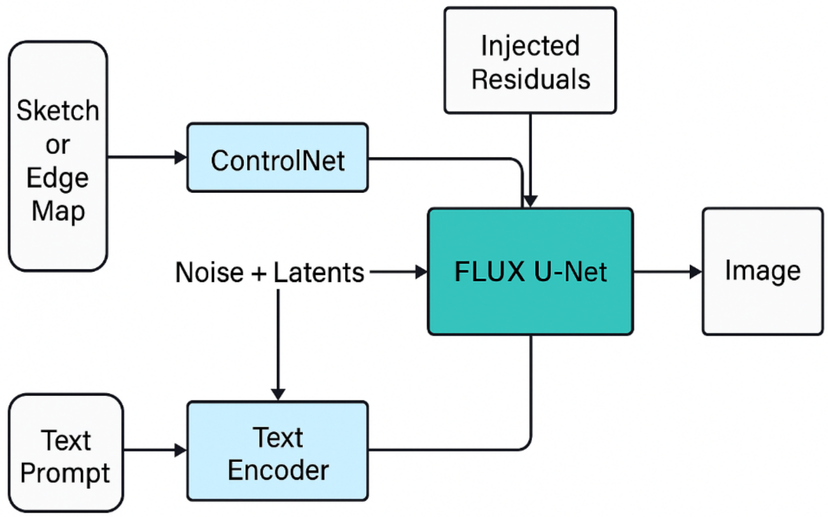
3. Noise-to-Image Generation with Semantic and Structural Fusion

- The FLUX U-Net operates in latent space, denoising random noise into an image while being continuously guided by:
    - Semantic features (from the prompt)
    - Structural cues (from the sketch)
- This results in high-quality images that look realistic, match the textual description, and respect the sketched layout—the central goal of Tracefy.

4. Improved Generalization and Robustness

- Thanks to its design, FLUX + ControlNet can handle a wide variety of face structures and descriptions. This robustness means Tracefy can support:
    - Abstract or rough sketches
    - Diverse facial features across ethnicities, ages, and styles
    - Complex prompts involving emotion, lighting, or accessories

13.2. Flux ControlNet Architecture



FLUX + ControlNet

The FLUX + ControlNet architecture is a dual-conditioning diffusion-based generative model designed to synthesize realistic images from both textual descriptions and structural inputs such as sketches or edge maps. It consists of three main components:

1. Text Encoder

- A pre-trained language model (e.g., CLIP or T5) is used to encode the text prompt.
- This produces a sequence of text embeddings that represent the semantic meaning of the user-provided description.
- These embeddings are later used in the cross-attention layers of the FLUX U-Net to guide generation.

2. ControlNet

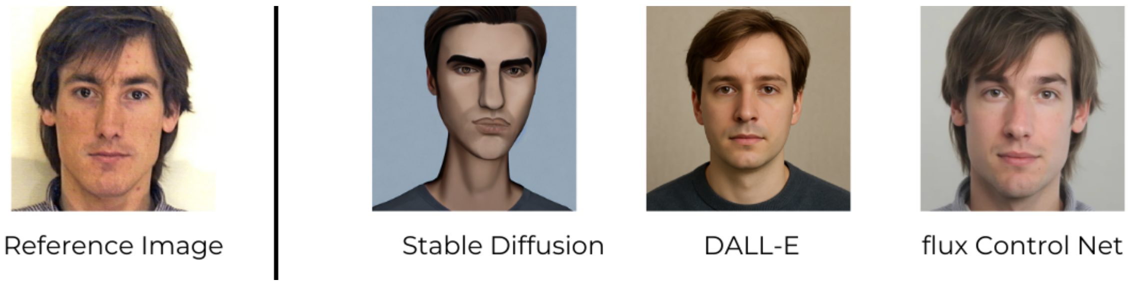
- ControlNet is a trainable neural network that receives the sketch or edge map as input.
- It mirrors the architecture of the FLUX U-Net's encoder and consists of convolutional blocks and residual connections.
- Rather than producing an image, ControlNet outputs intermediate feature maps (residuals) that capture the structural features of the input sketch.
- These residuals are injected into the FLUX U-Net at multiple depths, allowing the model to incorporate structural guidance throughout the denoising process.

3. FLUX U-Net (Modified UNet)

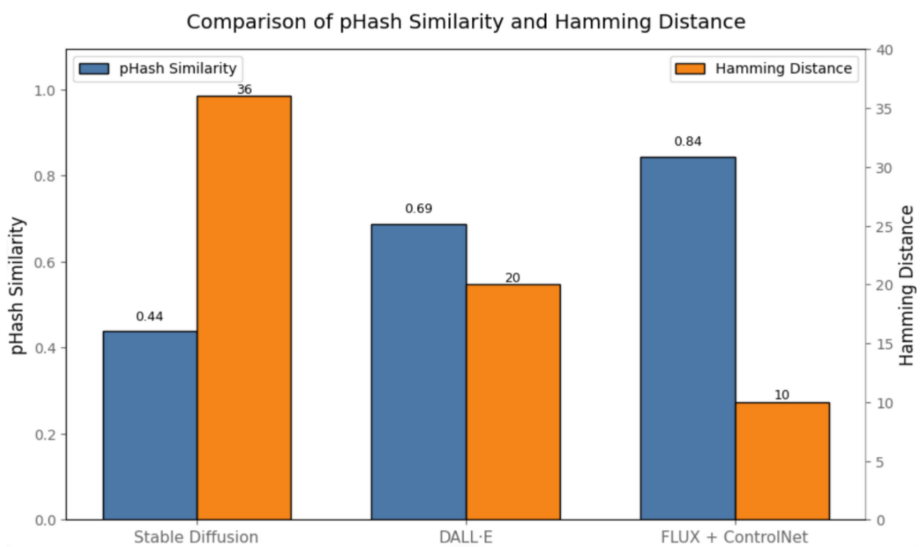
- The core of the architecture is a U-Net-based denoising network, adapted for latent space operations.
- It takes a noisy latent representation and gradually denoises it into a coherent image.
- The U-Net is guided by two streams:

- **Cross-Attention with Text**: At each attention layer, the model attends to the text embeddings, allowing semantic control over generated features such as gender, emotion, or hairstyle.
- **Structural Residuals from ControlNet:** At each layer, ControlNet injects feature maps that modulate the activations of the U-Net, ensuring structural consistency with the input sketch.

13.3. Results and Evaluation



The image above shows the difference in the results generated from three different models to compare their performance. Other than human evaluation, the numbers obtained by comparing the pHash Similarity and Hamming Distance shows that Flux-Control Net is best suited to perform the task of output image generation.



Based on the graph titled "Comparison of pHash Similarity and Hamming Distance", we can compare the performance of three face image generation models—Stable Diffusion, DALL·E, and FLUX + ControlNet—using two perceptual evaluation metrics: pHash Similarity and Hamming Distance. These metrics are crucial in assessing image fidelity in Tracefy's face generation pipeline.

Interpretation of Metrics:

- **pHash Similarity (left y-axis): A** higher value indicates greater visual similarity to the reference image.

- **Hamming Distance (right y-axis):** A lower value indicates less perceptual difference, hence better fidelity.

Model-wise Comparison:

1. Stable Diffusion

- **pHash Similarity**: 0.44 (lowest among the three)
- **Hamming Distance**: 36 (highest, indicating most dissimilar)
- **Inference**: Stable Diffusion performs the worst in terms of retaining facial identity features. This suggests its outputs are less faithful reconstructions of the input sketch/prompt.

2. DALL·E

- **pHash Similarity:** 0.69
- **Hamming Distance**: 20
- **Inference:** DALL·E shows moderate fidelity and perceptual similarity. It is a noticeable improvement over Stable Diffusion but not optimal for fine-grained identity preservation.

3. FLUX + ControlNet

- **pHash Similarity**: 0.84 (highest)
- **Hamming Distance**: 10 (lowest)
- **Inference**: This model combination offers the best results. The high similarity and low distance indicate strong alignment with the input identity—crucial for applications like Tracefy that rely on accurate face synthesis.
-

The **FLUX + ControlNet** pipeline significantly outperforms both **Stable Diffusion** and **DALL·E** in generating perceptually accurate face images. This supports its use in the **Tracefy system**, where maintaining identity fidelity from sketches and prompts is a key objective. The strong correlation of high pHash similarity and low Hamming distance further validates its robustness for trace-to-face generation tasks.

14. Future Work

Beyond its initial development stage, Tracefy's future is determined by a combination of immediate steps to follow, backup plans in case government adoption is slow, and a long-term expansion plan. Tracefy's B2G2C (Business-to-Government-to-Consumer) business model and digital architecture provide a solid basis for technical development, flexibility, and market expansion.

14.1. Technological Roadmap

Tracefy's roadmap includes a series of feature developments aimed at enhancing its functionality as a next-generation investigative platform. Each feature leverages cutting-edge AI/ML tools, cloud-native architectures, and privacy-preserving techniques:

- ***Voice-Based Prompting:*** Tracefy will utilize speech-to-text APIs such as Google Speech Recognition, Mozilla DeepSpeech, or Whisper by OpenAI, integrated with transformer-based NLP models (e.g., BERT, DistilBERT) to parse and understand spoken descriptions. These processed inputs will be passed to the sketch generation pipeline to render real-time visual outputs, reducing cognitive and physical barriers during high-stress interviews.

- ***Multi-Language Input Support:*** For inclusive usability across diverse linguistic regions in Pakistan, Tracefy will integrate multilingual transformer models such as mBERT, XLM-R, or IndicBERT for regional language comprehension. Backend support will leverage spaCy and Hugging Face Transformers for parsing natural language in Urdu, Punjabi, Pashto, Sindhi, and others, ensuring robust semantic mapping of descriptors to facial features.
- ***Courtroom Visualization & Crime Reconstruction:*** Future versions will include scenario simulation using 3D rendering engines like Unity3D or Blender, combined with OpenCV and GAN-based facial synthesis (e.g., StyleGAN2/3). These will enable visual reconstructions of suspects and crime scenes, exported as interactive modules or animated court exhibits. Scene elements will be composed using graph-based scene understanding models such as Scene Graph Generation (SGG).
- ***Advanced Model Optimization & Transfer Learning:*** To support efficient execution on edge devices, Tracefy will implement model pruning, quantization, and knowledge distillation techniques using frameworks like TensorRT, ONNX Runtime, and PyTorch Mobile. Transfer learning from base models such as ResNet, EfficientNet, or Vision Transformers (ViT) pre-trained on facial datasets will enable domain adaptation to regional traits with fewer data and computational resources.
- ***Secure Cloud Integration and Federated Learning***: Privacy-focused deployment will be enabled through federated learning frameworks such as TensorFlow Federated (TFF) and Flower, which allow decentralized training across user or institutional devices. Model weights will be aggregated centrally using secure multiparty computation (SMPC) or differential privacy mechanisms (e.g., via PySyft or Opacus) to maintain data confidentiality while ensuring performance.
- ***Explainable AI (XAI):*** Tracefy will integrate interpretability frameworks such as Captum (for PyTorch) or LIME/SHAP to produce saliency maps, attention heatmaps, and attribution graphs for the sketch generation model. For forensic accountability, these tools will highlight which textual descriptors most significantly influenced output features (e.g., eye shape, skin tone), ensuring transparency and aiding in legal admissibility.

14.2. Future Contingency Plan

Given the possibility of financial or bureaucratic delays in adoption by police departments, Tracefy has outlined multiple alternative use cases and entry pathways. These plans allow for broader deployment and early traction, ensuring the platform remains impactful and scalable even without immediate government partnerships.

14.2.1. Security and Civil Uses

- ***Corporate Security and Private Investigations:*** Tracefy can be deployed in corporate environments and private investigation firms as a forensic sketch generation tool powered by lightweight diffusion models such as Stable Diffusion or latent GANs. These models can operate on local servers or secure cloud infrastructure like AWS or Azure. A custom front-end interface—possibly built with React and Flask—can allow authorized personnel to generate suspect visuals based on structured text or guided voice input. This use is especially valuable in contexts where government database access (e.g., NADRA) is restricted.
- ***Missing Persons Support:*** Tracefy can serve NGOs and families by generating age-progressed or memory-based facial images using pretrained facial transformation pipelines. These models, implemented via PyTorch or TensorFlow Lite, can run efficiently on mobile devices and are designed for rural regions with limited connectivity. Features like gender/age sliders and regional facial template packs can improve visual accuracy.
- ***Fraud Detection and Identity Verification:*** Tracefy can enhance visual verification by integrating face descriptor models such as FaceNet or Dlib. It can be used as an additional step in digital KYC pipelines, matching descriptive inputs against a captured photo. Matching scores can be computed and indexed using tools like FAISS or We aviate for fast similarity retrieval while respecting data privacy through end-to-end encryption or federated model updates.

14.2.2. Self-Service App

- ***Community Safety and Watch Groups:*** Residents can create and share sketches of individuals they deem suspicious through integrated social sharing or API connections with local civic tech platforms. A metadata tagging layer could enable authorities to sort or flag submissions based on location, time, or keywords.
- ***Witness Convenience:*** Tracefy allows witnesses to create preliminary sketches from home using intuitive visual controls or voice-guided prompts. These can be exported (as PNG or JSON with descriptive metadata) and submitted to law enforcement through an official reporting portal, improving accuracy under stressful conditions.

14.3. Strategy for Scaling Tracefy

During a recent poster presentation, a National Incubation Centre representative supported this strategy, emphasising its potential to shift from a purely B2G model to a direct-to-consumer product. However, there may be ethical issues, especially regarding misidentification and misuse. Tracefy is designed to scale seamlessly in terms of both functionality and location. Rapid deployment with centralised updates is made possible by its cloud-based AI paradigm, which guarantees low infrastructure needs. Three strategic dimensions are considered in relation to scalability:

14.3.1. Geographic

Tracefy is deployable in rural police stations, provincial and federal agencies, and international law enforcement bodies—particularly in regions lacking forensic sketch artists or facing a rise in urban crime. By leveraging cloud services like AWS EC2 or Google Cloud Run, the system can be installed with minimal technical overhead, and updates can be pushed remotely. Its modular deployment model enables a plug-and-play setup, meaning even areas with limited infrastructure can access core features. For global expansion, multilingual and culturally adaptive sketch models—trained on diverse facial datasets—can be incorporated to ensure relevance in different countries

14.3.2. User

Beyond law enforcement, Tracefy is adaptable for private security firms, NGOs working in human trafficking and missing persons, and journalists investigating corruption or conflict. Granular role-based access controls (RBAC), powered by systems like Key cloak or Firebase Auth, will be integrated. This ensures tailored experiences for investigators, analysts, and external collaborators, limiting sensitive access to authorized personnel. Audit logs, secure sign-in, and OAuth-based identity verification will support trust and traceability in high-stakes environments.

14.3.3. Feature

The platform's modular architecture enables ongoing development. For example, to improve community reaction and engagement, alert and notification systems might be linked to public crime bulletins or missing child alerts. Enhancements to legal usage, including digitized simulations of suspect appearances or tools for visual storytelling in courtrooms, can improve the way legal procedures are presented and have an impact.

- ***Real-Time Collaboration & Versioning:*** WebSocket integration for multi-user collaboration on sketches and scene reconstruction, alongside version control of sketches and prompts, can enhance teamwork in large investigations.
- ***Predictive Modelling & Behavioral Forecasting:*** Additionally, using AI behavior models for predictive profiling may be able to predict a suspect's movements or changes in appearance over time, albeit such applications would need ethical review and permission.

14.4. Acquiring South Asian Dataset

To improve fairness, accuracy, and cultural specificity in sketch generation, Tracefy plans to acquire and curate a South Asian facial and sketch dataset. Current computer vision models often rely on datasets that are disproportionately Western in composition (e.g., CelebA, LFW), which can lead to reduced performance and biased outputs in Pakistani and broader South Asian contexts. Localized data is therefore essential for building equitable and effective AI systems.

- ***Data Collection Partnerships:*** Tracefy will collaborate with regional law enforcement agencies, NGOs, hospitals, and academic institutions to access anonymized real-world data. MOUs will be pursued with local universities and police departments to gather facial images, verbal descriptions, and sketches, under strict ethical oversight. Annotation pipelines can be developed using tools like Labelbox or CVAT to standardize and validate inputs.

- ***Ethical Guidelines:*** To maintain transparency and accountability, all data acquisition will follow documented consent protocols in line with international frameworks such as GDPR and Pakistan's Personal Data Protection Bill. An internal review board and audit process will oversee data use, storage, and access. Techniques like facial de-identification and differential privacy will be explored for sensitive data handling.
- ***Synthetic Data Augmentation:*** Generative models—such as StyleGAN2 or diffusion-based models fine-tuned on South Asian features—may be used to create synthetic faces and sketches, enriching the dataset while preserving privacy. These synthetic samples can be varied across attributes like age, gender, attire, and region, helping the model generalize better to real-world use cases without requiring excessive manual data collection.

14.5. Collaboration with organizations

Strategic collaborations are central to Tracefy's validation, growth, and impact. By engaging with key stakeholders across law, academia, civil society, and innovation ecosystems, Tracefy can strengthen its technical foundation, gain domain feedback, and accelerate real-world deployment.

- ***Law Enforcement & Justice Sector:*** Collaborations with police departments, forensic units, and public prosecutors will enable field testing, system validation, and iterative improvement of the sketch pipeline. Integration pilots may involve case simulation environments using local judicial workflows to assess admissibility and utility in court.
- ***Academic Research Labs:*** Joint R&D initiatives with AI and computer vision labs can enhance work on sketch-to-photo synthesis, adversarial robustness, and lightweight model deployment. Co-authored publications and shared compute resources (e.g., via Google Colab Pro or institutional clusters) can accelerate innovation while ensuring academic credibility.
- ***NGOs & Human Rights Groups:*** By working with civil society organizations active in anti-trafficking, missing persons, and refugee aid, Tracefy can be adapted for field use in humanitarian contexts. These groups also offer valuable feedback on ethical deployment, usability in underserved regions, and human-centered design practices.
- ***National Incubation Centers and Accelerators:*** Engagement with platforms like NIC Pakistan, NIC Karachi, or international accelerators such as Google for Startups will help refine Tracefy's business model, pricing structure, and legal strategy. These programs offer mentorship, regulatory pathways, cloud credits (e.g., through AWS Activate or Microsoft for Startups), and investor connections that can fast-track scale-up.

14.6. Pitching for startup

Tracefy is evolving into a robust AI startup with strong social impact and commercial viability. The pitch strategy focuses on key aspects that resonate with investors, partners, and stakeholders:

- ***Problem-Solution Fit:*** Emphasizing how Tracefy bridges the gap between outdated sketch-based crime solving and AI-generated facial reconstruction.
- ***Scalable Architecture:*** Showcasing its cloud-native design built on platforms like AWS or Azure, with a modular AI backend that enables multi-device deployment and supports multilingual interfaces using NLP frameworks such as Hugging Face transformers.
- ***Revenue Streams:*** Outlining diversified income channels including government contracts for law enforcement agencies, subscription-based API access tailored for private security firms, and a freemium self-service mobile app aimed at community users.
- ***Traction & Validation:*** Presenting proof-of-concept demos, user feedback from pilot tests and mock trials, as well as endorsements from innovation hubs like the National Incubation Center (NIC), reinforcing Tracefy's credibility and readiness.
- ***Ethical Considerations*****:** Emphasizing built-in safeguards such as explainable AI modules for transparency, compliance with responsible AI frameworks, and privacy-preserving technologies like federated learning to maintain data confidentiality.

15. Project Scope and Business Perspective

15.1. Scope

Tracefy is an AI-powered sketch generation tool designed to bridge the gap between manual criminal sketching and intelligent image synthesis. It converts structured prompts and input sketches into refined, photorealistic facial outputs. Built on a deep learning pipeline involving CycleGAN, Flux ControlNet, and LoRA fine-tuning, the system aims to improve identification accuracy, reduce sketching time, and eliminate human biases in visual reconstructions.

The current prototype includes the following functional components:

- ***Sketch-to-Face Generation Pipeline:*** Using a combination of Canny edge detection for preprocessing and CycleGAN for image-to-image translation, Tracefy transforms raw sketches into refined facial images.
- ***Prompt-Controlled Refinement:*** Flux ControlNet allows the user to influence output generation based on descriptive prompts. These inputs help tailor features such as hair type, facial structure, and eye shape, making the output more realistic and representative.
- ***Training and Fine-Tuning:*** The model has undergone LoRA (Low-Rank Adaptation) fine-tuning to improve convergence speed and efficiency, allowing quicker generation and domain adaptation for South Asian features.
- ***User Dashboard and Login System:*** A functional user interface enables input submission, sketch uploads, and output review in a secure and structured manner.
- ***Similarity Analysis Using pHash and Hamming Distance:*** To validate consistency and realism, perceptual hashing (pHash) is used to compare outputs to real sketches, quantifying similarity using Hamming distance.

15.2. Target Users

Tracefy is designed to serve a broad range of users across both public and private sectors, focusing on those involved in investigation, public safety, and visual identification. Its modular architecture and AI-driven functionality allow it to be tailored to diverse use cases—from national law enforcement operations to individual-level sketch creation.

Law Enforcement Agencies

The primary users of Tracefy are law enforcement personnel, including police officers, investigators, and forensic sketch teams. In many regions, particularly in rural or under-resourced areas, the absence of skilled sketch artists can delay or weaken investigations. Tracefy offers a scalable and accessible solution that enables officers to generate suspect visuals based on eyewitness descriptions quickly and with consistency. The system is especially useful during early-stage investigations when there is no photographic evidence available.

Security & Intelligence Organizations

Federal and provincial security bodies, anti-terrorism units, and intelligence agencies can use Tracefy to enhance profiling, suspect tracking, and cross-border crime investigations. By integrating structured descriptions with AI-generated imagery, Tracefy assists in situations where suspects are known only by verbal reports or partial recollections.

NGOs and Humanitarian Groups

Organizations involved in locating missing persons, combating human trafficking, or managing refugee crises can benefit from Tracefy's ability to generate visuals from memory-based inputs. This is particularly helpful in regions where official identity records or photographs are unavailable. The tool empowers social workers, volunteers, and affected families to contribute to visual identification efforts, even without technical expertise.

Journalists and Investigative Media

Reporters and investigative journalists working on stories related to corruption, crime, or conflict zones may use Tracefy to visually represent suspects or sources described by witnesses, without relying on unauthorized or manipulated photos. This ensures ethical reporting while maintaining anonymity where necessary.

Private Security Firms and Corporate Investigators

In the private sector, security consultants, investigative firms, and internal compliance teams can use Tracefy in corporate investigations, internal fraud detection, or surveillance follow-up. While these entities typically do not have access to national databases like NADRA, Tracefy offers them an independent, descriptive-based tool for visual documentation.

General Public (via Self-Service App)

A version of Tracefy tailored for public use can allow individuals to generate sketches for personal or community reasons. This includes members of neighborhood watch groups, users looking to reconnect with long-lost individuals, or witnesses who prefer to bring a pre-made sketch to a police station. The self-service model also aligns with growing public interest in generative AI technologies.

15.3. Stakeholder Analysis

Tracefy functions within a complex environment, instead of in isolation. The confidence and involvement of law enforcement agencies, witnesses, legislators, and the general public are essential to its success. Building trust, prioritising communication, and ensuring the solution's successful and long-lasting adoption in the public safety environment all required conducting a stakeholder analysis to identify and comprehend the various interests and impact of different groups.

Police Officers

Police have significant influence and interest in Tracefy since they are the main consumers. Their reliance on efficient, quick, and accurate suspect identification methods makes them critical stakeholders. Their input will influence how the technology is used and incorporated into regular tasks.

Government

Government agencies, particularly those in charge of justice, public safety, and technology, have a big say in procurement and budgetary choices. Even if they might not communicate with Tracefy on a daily basis, they still need to be satisfied and involved during financing and policy cycles.

Victims or Witnesses

These are indirect yet essential participants who provide suspect descriptions. Their role is substantial as they want prompt and equitable justice, despite their little authority. More accurate inputs and better results are guaranteed when the system is made accessible and easy to use for them.

Citizens

The broader public benefits from improved safety and quicker justice delivery. They are indirect beneficiaries with medium interest and limited influence. However, their trust in law enforcement and public discourse can shape support for AI in policing, so keeping them informed is key.

15.4. Business Model

Tracefy's business model is built upon a B2G2C (Business-to-Government-to-Citizen) structure, strategically aligned with the public sector's procurement and service delivery frameworks. This model enables Tracefy to offer its solution directly to government entities—specifically law enforcement agencies—who then utilize the product in their operations to ultimately benefit citizens. The B2G2C model leverages institutional purchasing power and regulatory authority while ensuring that the solution delivers a public good. Under this model, Tracefy is positioned as a technology vendor that provides its AI-powered identification software to police departments, investigation bureaus, and other criminal justice institutions through formal procurement channels. These institutions act as intermediaries that deploy Tracefy's solution in real-world crime-solving contexts, thereby indirectly serving the end beneficiaries: citizens seeking justice and safety. Tracefy ensures both financial sustainability and operational legitimacy by putting the government at the centre of the transaction, as institutional clients often enter long-term contracts, allocate money for technical modernisation, and have the capacity to mandate usage across departments.

Furthermore, this approach enables efficient scaling. Rather than targeting individual users or private security firms in isolation, Tracefy focuses on institutional adoption, which generates network effects across public sector ecosystems. As each new department or agency incorporates Tracefy into its crime investigation methods, the solution becomes more thoroughly embedded in the judicial system, promoting greater adoption across jurisdictions. Furthermore, the B2G2C model allows Tracefy to tailor its income streams to public sector procurement patterns, such as annual licensing, modular pricing depending on usage volume, or long-term integration contracts. The value chain is thus linear and controlled: Tracefy creates and maintains the product, sells it to government bodies, and then uses it to assist residents. This not only guarantees a mass effect, but also aligns with national priorities such as digitalisation, public safety, and organisational efficiency. Lastly, the B2G2C model offers a mission oriented and sustainable route to market, further cementing Tracefy's role as an essential tool in transforming Pakistan's criminal justice system.

Each of the business model canvas component's specifics are shown below:

Customer Segments

Tracefy's target segment is government departments that have a direct stake in public security, investigation, and citizen identification activities. Police organisations, both provincial and federal, and specialized investigative agencies like the Counter Terrorism Department (CTD) and the Federal Investigation Agency (FIA) are the main customers. These organizations are both the consumers and purchasers of the service, as they are charged with performing suspect identification for criminal cases. While Tracefy does not involve direct interactions with individual citizens, they are still indirectly served by the service since the final end of the model is to enable the government to better serve the public by accelerating and streamlining identification procedures. This framework ensures that the business model is institutional relationship-focused, supporting long-term interaction, high-value contracts, and organized integration into government systems.

Value Proposition

Tracefy's value proposition is in its ability to offer a revolutionary, high-precision identification solution that enhances the working efficiency of governmental agencies. Tracefy offers a cost-saving, time-efficient, and data-driven method of suspect identification that governmental agencies can utilize to accelerate investigations and reduce dependence on conventional sketch artists. Tracefy brings artificial intelligence to suspect identification, therefore addressing inefficiency and inaccuracy of manual means. The business model creates value by accelerating the law enforcement agency, improving precision, reducing backlogs in investigation, and delivering scalability across department and jurisdiction levels. In addition, the model includes ongoing enhancement through machine learning, enabling the system to adapt and enhance through every application, thus increasing value provided to government clients over a period of time.

Channels

Tracefy's communication and distribution channels are optimized for the B2G model. First, outreach is achieved through formal business development processes like government tenders, public sector procurement websites, and direct lobbying of ministries and departments responsible for law enforcement and public safety. Apart from formal channels, Tracefy can use pilot programs or Memoranda of Understanding (MoUs) with individual police departments to deliver proof of concept. Post-sale channels involve digital deployment, training of officers and administrators, and technical support. Tracefy can also utilize centralised government IT portals or regional digital transformation offices as middlemen to guarantee large-scale deployment. All communication and distribution channels are created to align with bureaucratic procedures without losing accessibility for end users—namely, government officials conducting criminal investigations.

Customer Relationships

In the B2G2C framework, Tracefy's customer interactions are mainly institutional and long-term. These relationships are formed and sustained through strategic partnership with government procurement organizations, technical support on a continuous basis, and value-added services like user training, integration customization, and system updates. Due to the nature of public sector customers, relationships are not transactional, but founded on trust, reliability, and persistent delivery of services. After being integrated into a law enforcement agency's process, Tracefy is a technology partner instead of merely a vendor. It is further reinforced by offering customer service portals, specialized account managers for government customers, and the introduction of feedback loops that enable agencies to provide feedback on system performance.

Key Resources

Tracefy's business operations are critically reliant on an integration of technological, human, and institutional resources. The main technology resources are specialized AI algorithms, cloud-based data processing and storage infrastructure, and secure integration technologies enabling real-time data exchange with national databases such as NADRA.

Key Activities

Tracefy is involved in numerous key activities such as developing products, client interaction, and post-sales service to ensure its value proposition is effectively delivered. Some of these activities include AI model continual improvement to enhance identification accuracy, software testing and quality, interaction with government databases, and jurisdictional customizations. Business development activities like proposal draughting, government tender bidding, and technical demo provision are also equally important. Tracefy's post-sale process involves onboarding client teams by conducting training programs, resolving user questions, providing real-time assistance, and publishing system updates due to client comments.

Key Partnerships

Tracefy's success is dependent on significant partnerships in technological, institutional, and governmental sectors. The most significant collaboration is with NADRA for future database integration, allowing identity matches for suspect photos. Collaboration with law enforcement organisations, particularly the Islamabad and Peshawar police, is essential for pilot testing and user feedback. Future collaborations with legal experts and public sector innovation hubs will also help Tracefy grow while adhering to data regulations and procurement protocols.

Cost Structure

Tracefy's high expenses are cloud storage and GPU processing, which are used to execute the AI system. The technical development team salaries contribute to the overall expenses. As the system expands, other expenses will be server infrastructure, training, and compliance legal. Generally, the cost structure is set to maintain and enhance a technology-based solution for public sector utilization.

Revenue Streams

Tracefy earns revenue based on charging the police departments a fee per sketch that is completed. Tracefy has a tiered pricing scale based on service quality. Tracefy also sells annual service plans for technical maintenance and upgrading. Tracefy can feasibly offer provincial or federal government bulk licensing in a way that enables wider use across several departments. Tracefy will offer usage volume discounts for heavy usage quantity while guaranteeing long-term top-line revenue growth. Tracefy will remain economical, profitable, and scalable under Pakistan's public sector model through the incorporation of usage-based pricing and recurring service packages.

Key Partners

- NADRA (planned database integration)
- Police departments (pilot testing and implementation)
- CS students (technical development team)
- Legal advisors (for compliance and contracts)
- Public innovation hubs and accelerators (for funding/support)

Key Activities

- AI model development and updates
- Database integration and system scaling
- User onboarding and training
- Responding to government tenders
- Post-sale support and feedback analysis

Key Resources

- AI models and backend architecture
- Google Cloud storage and GPU compute systems
- Technical team: 3 CS students (AI training, system design, backend)
- Business team: financials, partnerships, stakeholder coordination

Value Propostition

- Fast and accurate AI-generated suspect sketches
- Integration with NADRA for real-time identity matching
- Reduces reliance on traditional sketch artists
- Speeds up investigations and reduces backlog
- Scalable system that learns and improves over time

Customer Relationships

- Long-term institutional contracts
- Dedicated technical support for each department
- Custom training sessions for officers
- Feedback integration for system improvement
- Maintenance agreements and annual service contracts

Channels

- Government tenders and procurement platforms
- Direct proposals to law enforcement and government departments
- Pilot programs and early adoption partnerships
- Workshops, demos, and onboarding sessions

Customer Segments

- Government law enforcement agencies (Police, CTD, FIA)
- Metropolitan police departments (e.g., Islamabad, Peshawar)
- Indirect beneficiaries: citizens (victims/witnesses)

Cost Structure

- Cloud storage and GPU compute costs (monthly tech expenses)
- Salaries of the technical development team
- Future costs for server infrastructure, legal compliance, and training
- Scalable costs as user base and data grow

Revenue Streams

- Per-sketch pricing (tiered: Basic, Standard, Premium)
- Annual maintenance and technical support contracts
- Bulk discounts and province-wide licensing opportunities
- Future potential: analytics and dashboard add-ons

Business Model Canvas

16. Conclusion

Tracefy represents a confluence of cutting-edge technologies in generative AI, human-computer interaction, and digital creativity. Technically, it demonstrates the viability of a multi-stage sketch-to-photo generation pipeline powered by CycleGAN for image-to-image translation and FLUX.1-dev's latent diffusion framework fine-tuned via LoRA. The addition of ControlNet ensures structural fidelity to user sketches, while text prompts guide nuanced semantic interpretation. The modular system design enables clear abstraction between components, facilitating scalability and experimentation.

Our model evaluation strategy ensures both qualitative and quantitative rigor through perceptual and structural similarity metrics. The user interface successfully abstracts the complexity of the backend models, allowing users to interactively create, refine, and visualize facial generations with ease. Design models and UI components are tailored to provide real-time feedback, ensuring responsiveness and creative control.

From a business and product perspective, Tracefy opens new avenues in content creation, character design, and digital forensics, supporting a wide range of stakeholders from artists and animators to security personnel. The system's architecture is designed with extensibility in mind, enabling future upgrades such as multimodal input (voice or reference images), zero-shot learning for style transfers, and personalized model fine-tuning per user profile.

Future work will focus on enhancing generation speed and quality, improving fine-tuning workflows, and deploying lightweight models for edge devices. Additional integration with AR/VR environments and expansion into full-body generation are also envisioned. Overall, Tracefy contributes meaningfully to the field of generative AI by bridging human creativity with machine precision.

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18. Glossary

1. **Sketch Generation**: The process of creating visual representations (sketches) based on input data, often used in law enforcement for identifying suspects.

2. **User Roles and Permissions**: Defines different levels of access within the Tracefy system, such as Law Enforcement Officers, Administrators, Investigative Analysts, and External Consultants.
3. **Bias Mitigation**: Techniques implemented in the Tracefy system to reduce or eliminate biases in AI-generated sketches, ensuring fairness across different demographics.
4. **Feedback Mechanism**: A system that allows users to provide input on the generated sketches, which can be used to improve future model predictions.
5. **Sketch to Sketch**: A process in the model pipeline where one sketch is transformed into another, enhancing the quality and accuracy of the generated images.
6. **Sketch to Face**: The final stage in the model pipeline where a sketch is converted into a realistic facial representation, providing a usable output for law enforcement.
7. **Real-Time Features**: Functionalities that allow the Tracefy system to process and respond to user inputs and actions instantaneously, enhancing user experience.
8. **B2G2C (Business-to-Government-to-Citizen)**
   A business model where a company provides services or products to a government entity, which then uses those solutions to serve citizens.
9. **Procurement**
   The formal process by which government agencies acquire goods and services from external vendors.
10. **Institutional Purchasing Power**
    The ability of large organizations, especially governments, to make high-volume or long-term purchases.
11. **Regulatory Authority**
    The legal power granted to institutions, especially government bodies, to enforce laws or regulations.
12. **Technology Vendor**
    A business that supplies technological products or services to other organizations.
13. **Intermediaries**
    Entities (such as government bodies) that facilitate the delivery of services from a business to the end user.
14. **Pilot Programs**
    Small-scale, preliminary studies used to evaluate feasibility, time, cost, risk, and performance before a full-scale rollout.
15. **Memoranda of Understanding (MoUs)**
    Formal agreements between parties that outline mutual intentions and terms for collaboration.
16. **Public Sector Procurement**
    The acquisition of goods, services, or works by government institutions, typically through competitive bidding or tender processes.
17. **System Integration**
    The process of linking different computing systems and software applications physically or functionally to act as a coordinated whole.
18. **Digital Transformation**
    The integration of digital technology into all areas of a business or public service, fundamentally changing how it operates and delivers value.

19. **Cloud-Based Infrastructure**
    Technology services (such as servers, storage, and databases) hosted remotely and accessed over the internet.
20. **GPU Processing**
    Use of Graphics Processing Units to accelerate tasks involving large-scale computation, particularly common in AI/ML applications.
21. **NADRA (National Database and Registration Authority)**
    Pakistan's national institution responsible for citizen identity management.
22. **Usage-Based Pricing**
    A revenue model where charges are based on the amount a service is used rather than a flat fee.
23. **Bulk Licensing**
    An agreement that allows multiple users within an organization to use software under a single license contract.
24. **Jurisdictional Customization**
    Adjustments made to a product or service to comply with or serve specific legal, administrative, or geographic boundaries.
25. **Law Enforcement Agencies (LEAs)**
    Government bodies responsible for the enforcement of laws and maintenance of public order and safety.
26. **Revenue**
    The total income generated from the sale of goods or services before any costs or expenses are deducted.
27. **Recurring Revenue**
    Predictable and consistent income generated at regular intervals, typically from subscriptions or service contracts.
28. **Technical Maintenance and Upgrading**
    Ongoing processes to ensure that software systems remain functional, secure, and up-to-date.
29. **Feedback Loops**
    Systems in which outputs of a process are used as inputs for future iterations to improve performance or accuracy.
30. **Onboarding**
    The process of introducing and integrating a new client or user into a product or service.